

When a Protectionist Trade Policy turns into an Industrial Policy: Evidence from India*

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Abstract

Technical Barriers to Trade (TBTs) can serve as powerful protectionist tools, yet it remains unclear whether they benefit domestic firms. This paper studies India's 2001 regulation mandating testing and certification for vehicle components. Using detailed firm-level data (1997–2005) and exogenous variation in exposure, we document three main effects for import-dependent automotive firms: (i) increased technology adoption, driven by R&D and technology transfer; (ii) upgrading of input mix toward higher-quality inputs while maintaining product lines at higher value; and (iii) gains in market share, export expansion, and higher Revealed Comparative Advantage. We underscore that TBTs can act as de-facto industrial policy.

Keywords: TBTs, Industrial policy, Technology adoption, Input composition, Product portfolio, Export development

JEL Codes: F13, F14, O25

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1 Introduction

Technical Barriers to Trade (TBTs)—product standards, testing requirements, and certification procedures—have become a dominant instrument of modern trade policy.¹ TBTs now account for a majority of new trade concerns raised at the WTO ([World Trade Organization, 2025](#)), with developed and developing economies alike increasingly relying on regulatory measures to govern market access. Unlike tariffs, which regulate *how much* countries trade, TBTs regulate *how* firms produce. This dual nature—trade barrier and regulatory mandate—means that TBTs, while restricting foreign competition, can simultaneously reshape the cost structure and capabilities of domestic firms, effectively functioning as *de facto* industrial policy.²

This paper investigates whether TBTs can generate industrial upgrading in the domestic economy. We study a set of technical regulations imposed by India in 2001 that targeted the automotive sector, requiring mandatory testing and certification for all new vehicles and components by the Automotive Research Association of India (ARAI). Using detailed firm, firm-input, and firm-product level data, we estimate the causal effect of these regulations on technology adoption, input quality, product portfolios, and export performance. We find that TBTs can act as *de facto* industrial policy, upgrade production technology, and export competitiveness. This is our primary contribution.

India provides a compelling setting. Following decades of import substitution, the country undertook major trade liberalization in the 1990s under IMF-conditioned structural adjustment.³ However, from the early 2000s, the pace of tariff reform slowed and trade policy increasingly relied on TBTs to regulate market access. This evolution in India’s trade policy is particularly relevant for our analysis of TBT or technical regulations. Not all such regulations are trade-restrictive: some address information asymmetries ([Moenius, 2004](#)) or facilitate coordination ([Fontagné et al., 2005](#)). To isolate genuinely trade-distorting measures, we draw on the WTO’s *Specific Trade Concerns* (STCs) database, which records measures formally challenged by member states as unnecessary obstacles to trade. **Figure 1** shows that 32 such concerns were raised against India between 1995 and 2020, beginning in 2001.

Unlike tariffs, technical regulations combine two features that can pull domestic investment incentives in different directions. On one side, they can restrict imports and

¹TBTs, which are a type or form of NTBs (Non-tariff Barriers) are usually implemented to pursue legitimate objectives – such as consumer safety, product quality, environmental, and/or health protection, etc. However, they often carry implications that extend beyond simple trade regulation.

²This interpretation is consistent with strategic trade and industrial organization models ([Brander & Spencer, 1985](#); [Baldwin, 1989](#)).

³In 1991, India turned to the IMF following a balance-of-payments crisis. Assistance was conditioned on a broad adjustment program that included abandoning restrictive trade policies. For details, see [Chakraborty & Raveh \(2018\)](#).

reduce foreign competitive pressure. On the other, they can impose mandatory compliance requirements—testing, certification, redesign, and process adaptation—that raise the need to invest in production capabilities (Chakraborty, 2017; Hu *et al.*, 2019; Wei *et al.*, 2023). The first force is often associated with weaker innovation incentives in the infant-industry critique (Tybout, 2003; Slaughter, 2004). Conversely, if firms perceive regulation(s) as a credible threat for their operations in the domestic market and anticipate future competition, they may respond proactively by investing in new capabilities (Lane, 2025; Giorcelli, 2019; Rotemberg, 2019; Martin *et al.*, 2017).⁴

For our purpose, we exploit the STCs that are classified as TBTs and raised by WTO member countries against India between 1997 and 2005. In particular, we label the TBTs into two groups: (i) retail regulations – regulations that introduced specific labeling requirements and shelf life lower than a certain threshold. These regulations covered a wide set of perishable (especially food and beverages) and nonperishable products for retail consumption; and (ii) automotive regulations – which targeted the automotive industry and introduced new mandatory tests on all new vehicles and components with a certification from the Automotive Research Association of India (ARAI).

Empirically, we estimate a difference-in-differences design on PROWESS firm-, firm-input-, and firm-product level data for 1997–2005. We exploit variation in compliance intensity by comparing high-compliance automotive regulations (STCs 84 and 104) with lower-compliance retail regulations (STCs 54 and 70), while also using untreated firms as an additional comparison group.⁵ This design allows us to test whether upgrading is primarily associated with compliance intensity rather than with trade protection alone.

Our framework therefore distinguishes between two channels. The first is *competitive insulation*: by restricting imports, TBTs can relax competitive pressure and potentially weaken incentives to invest. The second is *compliance-driven investment*: mandatory testing and certification can directly increase the return to capability accumulation, including testing infrastructure, technology adoption, and process upgrading, reflected in R&D and technology transfer expenditures. The central empirical question is which force dominates in this setting. If insulation is the dominant margin, effects should mainly reflect protection. If compliance intensity is dominant, we should observe stronger upgrading precisely where certification requirements are more binding, including higher-quality input use, production upgrading, and potentially stronger market performance.

⁴Historical experience provides both sides of the story: while many protectionist interventions failed, the U.S. automotive industry in the 1980s is often cited as a case where temporary trade barriers—through voluntary export restraints—helped stabilize employment and spur investments in product quality (Feenstra, 1995).

⁵The comparison group therefore combines firms exposed only to retail-sector TBTs and firms not exposed to any TBT.

Our findings document a policy-induced cascade from trade margins to firm capabilities and market outcomes. First, the automotive TBTs reduce imports in targeted categories, indicating that the policy meaningfully altered competitive conditions at the border. Second, firms in exposed automotive industries respond by upgrading capabilities: technology-adoption expenditures rise by about 38%, with significant increases in both R&D (about 23%) and technology transfer (about 30%); input bundles shift toward higher-value components (average input expenses rise by about 24%); and product portfolios consolidate rather than diversify. Third, these production-side adjustments are accompanied by economically meaningful market outcomes, including higher domestic sales (about 30%), stronger export performance on both intensive and extensive margins, and improved automotive export competitiveness at the aggregate level.

To identify the source of these effects, we leverage within-variation in the automotive regulatory bundle and isolate its distinct components, exploiting the fact that an additional import ban applies only to a subset of products. This allows us to assess whether the observed upgrading is primarily driven by compliance requirements rather than by import restriction alone. We then show that responses are strongest among firms with greater pre-shock import dependence and are accompanied by reallocation away from imported capital-goods reliance toward domestic technology transfer and R&D capital. Taken together, the evidence is most consistent with a compliance-driven capability-upgrading channel, with import exposure shaping the intensity of adjustment.⁶

These results make two important contributions. First, we provide new evidence that non-tariff measures can operate as a contemporary form of industrial policy. The existing trade literature has documented how TBTs affect trade volumes (Shephard, 2007; Bao & Qiu, 2012; Fontagné & Orefice, 2018), markups (Singh & Chanda, 2021), and organizational structure (Navaretti *et al.*, 2024); we show that TBTs can fundamentally reshape domestic production processes through technology adoption, input upgrading, and export development. This aligns with the emerging view that industrial policy encompasses a broad range of government interventions shaping sectoral composition, beyond direct subsidies or tariffs (Juhász *et al.*, 2024).

Second, our findings speak to the infant-industry debate. Classic infant-industry episodes involve *deliberate* temporary protection designed to build capabilities; the TBTs we study were regulatory requirements not explicitly intended as developmental tools, making the observed upgrading an unintended byproduct. That compliance-driven investment, rather than import restriction, generated the upgrading offers a new mechanism through which trade regulation can foster capability accumulation—distinct

⁶We complement our firm level analysis with a back-of-envelope welfare assessment that quantifies static consumer costs relative to producer gains.

from both traditional protection and the input-trade liberalization channels documented by [Amiti & Konings \(2007\)](#), [Goldberg *et al.* \(2010\)](#), and [Bas & Strauss-Kahn \(2015\)](#). Notably, while [Goldberg *et al.* \(2010\)](#) find that input tariff reductions in India led firms to introduce new products, we observe consolidation rather than diversification—consistent with regulatory incentives to deepen existing capabilities rather than expand product scope.

Our work also connects to recent quasi-experimental studies of industrial policy. [Lane \(2025\)](#) documents persistent productivity gains from Korea’s heavy and chemical industry drive; [Giorcelli \(2019\)](#) shows durable effects of U.S. management transfers to Italian firms; [Juhász \(2018\)](#) finds that Napoleonic-era trade barriers fostered mechanized cotton production in France. Similarly, [Rotemberg \(2019\)](#) and [Martin *et al.* \(2017\)](#) document heterogeneous effects of India’s small and medium enterprise programs, showing that targeted support can increase productivity and sales among eligible firms. On the theory side, early contributions emphasized coordinated state intervention to overcome market failures ([Rosenstein-Rodan, 1943](#); [Nurkse, 1953](#); [Hirschman, 1958](#)), while the postwar East Asian experience provided influential examples of state-led industrialization ([Amsden, 1992](#); [Wade, 1990](#)). Our paper shows that regulatory trade barriers—operating through compliance mandates rather than fiscal transfers—can produce comparable upgrading effects.

The remainder of the paper is organized as follows. Section 2 describes the data, the institutional background, and presents some stylized facts. Section 3 presents the empirical strategy, discusses the issues with the estimation strategy, and examines the exogeneity of the policy change. Section 4 estimates first-order trade effects in terms of import flows. Section 5 reports the effect on production capabilities in terms of technology adoption, input mix, and product portfolio. Section 6 explores what happened to sales, export probabilities, and revealed comparative advantage with a back-of-the-envelope welfare assessment. Section 7 concludes.

2 Data

We combine data from three different sources to analyze the impact of TBTs on different outcome measures of Indian manufacturing firms. In particular, we draw data from the following sources: (1) WTO’s STC database, which captures each country’s concerns about potentially trade-restrictive measures imposed by foreign governments, specifically their trading (say, importing) partners; (2) PROWESS database – a rich source of financial and production level information on Indian registered manufacturing firms; (3) product level (6-digit HS Code) trade data from the UN COMTRADE database,

maintained by the World Integrated Trade Solution (WITS). Below, we describe each dataset in detail.

2.1 Trade Concerns Database

The STC dataset provides information on: (i) WTO member(s) raising concerns regarding a trade measure and the country who maintains that measure, (ii) the date on which the concern is raised and all subsequent dates when the concern is further discussed in the committee, (iii) the 4-digit HS Code of the product(s) targeted by a specific trade concern, in our case a TBT, (iv) the objectives of the concern, and (v) lastly, the concern(s) that a country or countries raise in regard to the specific trade measure.

In essence, detailed information on each STC allows us to identify the measures that other countries have explicitly signaled as potentially trade distorting. In the case of India, foreign countries raised 32 concerns between 1995 and 2020 with the first concern dating back to 2001. We focus only on those concerns which are classified as TBTs. Following [Singh & Chanda \(2021\)](#), we consult the relevant notifications in the Gazette of India to retrieve all the dates when TBTs were effectively implemented.⁷

For our purpose, we focus on those STCs which were introduced before 2005. This would allow us to minimize concerns about (a) the higher frequency of TBTs after 2008 which could be the result of the 2008 financial crisis ([Kee et al., 2013](#)) rather than other concerns about policy implementation; and (b) the overlap of treatment and/or anticipation effects, thus making it easier to isolate the causal impact of TBTs on domestic firms. The details of the TBTs/STCs that we exploit in our study, the underlying regulations' content, timing are outlined in **Table 1**.

In particular, our study exploits the STCs which are numbered 54, 70, 84, and 104.⁸ Based on the content and industry coverage of these TBTs, we divide the regulations into two groups: (i) regulations 54 and 70, which introduced specific labeling requirements, as well as bans for products with remaining shelf life lower than a certain threshold. These two regulations covered a wide set of perishable and nonperishable products for retail consumption, in particular those belonging to the food and beverages sector. We will refer to these regulations as *retail* regulations;

⁷In essence, the STCs dataset does not provide information on the date of implementation of the contested measures, which can be different from the year on which the concern was raised. Concerns may be raised in a preliminary way (i.e., before the formal implementation), in the same year, or after a TBT enters in force. In some cases, they address regulations that never came into force. Thus, we retrieved the dates on which the TBTs were effectively implemented based on the notifications made in the Gazette of India.

⁸For brevity, we refer to each measure by its STC number. Regulation 55 is excluded as it covered only HS codes 2207 and 2208, for which we have no firms in the dataset.

(ii) on the other hand, regulations 84 and 104 targeted the automotive industry and entailed a set of stringent requirements. Specifically, the regulations introduced new mandatory tests on all new vehicles and components, which required a certification from the ARAI. Additionally, ARAI did not have testing capabilities for large engines, so foreign firms had to set up their own facilities or use the facilities of those firms which had similar set up. Finally, these regulations also banned the import of second-hand vehicles older than three years and imposed a special certification for the younger ones. We label these regulations as *automotive* regulations.⁹

2.2 Firm level Data: PROWESS

Our firm level data comes from the widely used PROWESS database provided by the Centre for Monitoring the Indian Economy (CMIE). The dataset reports firm level information from income statements and balance sheets for publicly listed firms, and from yearly surveys of unlisted firms. The firms in the dataset cover more than 70% of economic activity in the organized industrial sector, and 75% (95%) of corporate (excise duty) taxes collected by the Indian Government. The majority of the firms in the dataset are either private Indian firms or affiliated to private business groups, whereas a small percentage of firms are government or foreign-owned. Firms in the dataset span 116 disaggregated manufacturing industries (at the 4-digit National Industrial Classification (NIC)), and has been widely used in studies focusing on firm outcomes in India (see for instance, [Topalova & Khandelwal \(2011\)](#); [Chakraborty & Raveh \(2018\)](#)). We use data for about 6000 firms.

For our purpose, we use data at two different levels: (i) information on firm-year variables from the balance sheet of the firms. One of our key outcome variables that we use to investigate the effect of the concerned TBTs (i.e., the *Automotive* regulations) is the effect on the R&D expenses and expenses on account of foreign and domestic technology transfer. According to this dataset, firms must report information on R&D expenditure (further divided into capital and current) and technology transfer as per section 217 of the Companies Act. i.e., as per section 217(1)(e), the information shall be attached to every balance sheet laid before a firm in the Annual General Meeting, in a report by its board of directors.

We also use data on total sales, domestic sales, raw material expenses, exports, imports – further disaggregated into raw materials, capital goods, stores and spares, and final goods, assets, ownership category, establishment year, etc.

and (ii) details on the input used and products produced by a firm. Following [Vandenbussche & Viegelaan \(2018\)](#), we use detailed data on firm-input that hitherto has

⁹We provide all the details regarding these regulations in **Appendix A**.

not been explored much in the literature. The dataset provides disaggregated information on the value and quantity of different raw material inputs that the Indian manufacturing firms are using in production. A key ingredient of the *Automotive* regulations is that it mandated firms to get a certification from ARAI to use new components and vehicles. This aspect of the regulation can significantly impact their choice of inputs they use to produce new products.

2.2.1 Stylized Facts

Fact # 1. *Firms treated with automotive TBTs use inputs of higher value than others.*

Panels A and B of Table C.1 present a few summary statistics regarding the number of inputs used and the value of inputs used for firms classified into four different groups: aggregate, firms belonging to industries that have been treated with automotive TBTs, firms belonging to industries that have been treated with retail TBTs, and firms which were not exposed to any TBT.

A firm belonging to an industry treated with automotive TBT uses on average 4 inputs. This is similar for firms that correspond to industries with retail TBTs or firms that have been treated with no TBTs. Interestingly, in terms of the value of inputs used, a firm that has been treated with an automotive industry TBT uses input value which is 1.5 times the value used by an average firm treated with a retail TBT and more than double that of the firms that have not been exposed to any TBT. This indicates that a firm that has been exposed to an automotive TBT uses inputs which are on average of higher value or higher quality.

Fact # 2. *Firms treated with automotive TBTs also produce products of higher value than others.*

Similarly, the regulation(s) can also impact the products produced by each firm. For example, firms may introduce new products and drop old or existing ones thereby resulting in a significant product churning. On the other hand, drop in competition (due to reduction in imports) may compel them not to shed, but to produce the existing products in their product portfolio with better quality inputs than before, therefore having little effect on product churning. PROWESS tracks a firm's product mix over time. This is primarily due to the Companies Act of 1956, which requires Indian firms to disclose product level information on capacities, production, and sales in their annual reports. [Goldberg et al. \(2010\)](#) highlight that product level information is available for 85% of manufacturing firms, which in turn accounts for more than 90% of PROWESS' manufacturing output and exports. PROWESS reports data on sales (in Indian Rupees) and quantity sold at the product level, classified according to CMIE's internal product

codes. The vast majority of firms in our sample produce two or more products.¹⁰

Panels C and D of Table C.1 show similar trends in terms of the number of products and the value of products produced. Whilst there are no substantial differences in the number of products produced between treated and non-treated firms in our sample, automotive firms stand out with a substantially higher average product value.¹¹

2.3 International Trade Data

Lastly, we use product level (at the HS code) import data from the UN-COMTRADE database, accessed via the World Bank’s World Integrated Trade Solution (WITS). COMTRADE provides detailed data on merchandise imports and exports, recorded at the HS 4-digit level, for over 170 countries. For our purpose, we extract yearly data on Indian imports disaggregated by product category and partner country. These product-country level imports data would allow us to estimate how sector level imports respond to the imposition of TBTs. The use of this HS 4-digit country-product level trade data complements our firm level analysis by providing a broader picture in terms of understanding our first order effects, i.e., trade patterns of those industries where the TBTs have been imposed.

3 Empirical Strategy

To assess the impact of the concerned TBTs on firm behavior, we follow a simple difference-in-differences (DiD) framework where we use firms belonging to industries affected by regulations 84 and 104 as the treated group of firms and compare them with firms that are affected by regulations 54 and 70 and those unaffected by any TBTs, i.e., firms which form our control group. We use the following baseline model:

$$\ln(x_{ijt}) = \beta_1 \text{Treated}_{jt} + \beta_2 (\text{Treated}_{jt} \times \text{Automotive}_{ij}) + \alpha_i + \gamma_t + \epsilon_{ijt} \quad (1)$$

where x assumes different firm level outcomes for a firm i belonging to industry j at time t , such as total expenses on innovation which is a sum of R&D expenditure (capital and current) and expenses on account of technology transfer (both foreign and domestic), total sales, domestic sales, exports.

The treatment indicator Treated_{jt} is the interaction of two terms: (i) Post_t , a year dummy variable that takes value 1 in years after 2001 and 0 otherwise; and

¹⁰Only 656 out of 4643 (14%) reports an average of one product during the years of analysis.

¹¹Details of all the variables used in our analysis are laid out in Appendix B. We also present the summary statistics for all the variables used in Table C.2.

(ii) *Treated Industry_j*, which takes value 1 for industries (*j*) exposed to any of the Regulations 54, 70, 84, or 104. We define those industries to be treated if they are operating in 4-digit product categories (HS codes) explicitly targeted by TBT-related trade concerns. To assign industries to these groups, we first map the STC dataset's HS 2002 product classifications to NIC 2008 codes as reported in PROWESS against each firm's industry classification. Because HS codes correspond more closely to ISIC Rev. 3 (equivalent to NIC 2004) than to NIC 2008, we first convert NIC 2008 codes to NIC 2004 using an official concordance table from the UN and MoSPI (Ministry of Statistics & Program Implementation, Government of India).¹² Through this procedure, we mapped the full set of 1,244 HS 4-digit codes in HS 2002 to 615 NIC 2004 4-digit codes. Since the focus is manufacturing firms, we restrict our attention only to manufacturing codes (145).

Industries linked to treated HS 4-digit codes via NIC 2004 are then considered to be treated if any of the corresponding HS codes are covered by a TBT, with treatment assigned from the earliest year of concern.¹³ Our final sample consists of an unbalanced panel of 4,995 firms, belonging to 99 4-digit different NIC 2004 categories.¹⁴

Our matching of HS Codes with industry codes results in two sets of firms – firms treated with TBTs and firms without TBTs. Now, our identification relies on the assumption that, in the absence of treatment, firms in treated and control (i.e., firms with no TBTs) industries would have followed similar trends in their outcomes. This parallel trends assumption is crucial for the validity of our difference-in-differences approach and cannot be directly tested. **Table C.4** run balance tests on these two sets of firms to check whether in the pre-policy period, firms differed across the key outcome variables or not. In particular, we compare the pre-treatment (1997–2001) averages for a number of firm level characteristics, such as R&D, technology transfer expenses, sales, exports, assets, age. The normalized differences show that firms did not vary across these observables in the pre-policy period.¹⁵

¹²There are cases when a single NIC 2008 code mapped into multiple NIC 2004 codes. In those cases, we retained only those matches where the resulting codes share the same first three digits, indicating close industrial similarity.

¹³As can be understood, this procedure may introduce some conservative bias due to potential misclassification as the matching between NIC 2008 and 2004 is done based on a firm's activity and main production group. However, we also tried a few alternative ways of matching, such as by dropping all the ambiguous matching pairs of matching between NIC 2008 and 2004 and keeping just the exact matched pairs or keeping only those 4-digit NIC codes which are highly likely to be effectively treated, by establishing a threshold. All of these procedures yield very similar results.

¹⁴**Table C.3** summarizes the number of industries at 4-digit NIC level that belong to the various combinations of treatment and control groups. Overall, 67% of the sample is subject to one or more TBT-related regulations, the majority of which fall under the retail regulations, while the industries which were affected by the automotive regulations account for only 12% of the treated group.

¹⁵In following exercises, we provide event-study specifications for every key outcome variables of interest, where we interact year fixed effects with the treatment indicator and verify that pre-treatment coefficients are close to zero and jointly insignificant.

We then interact $Treated_{jt}$ with $Automotive_{ij}$, an indicator equal to one if a firm i operates in an industry j targeted by the automotive regulations (84 and 104). The coefficient β_1 captures the effect of low-compliance retail TBTs (Regulations 54 and 70) relative to untreated firms. And, the coefficient β_2 captures the additional high-compliance automotive effect. Therefore, $\beta_1 + \beta_2$ is the total effect for automotive firms, which we report throughout.¹⁶

This design serves a dual purpose. Regulations 54 and 70 primarily imposed labeling and shelf-life requirements with low compliance costs, while automotive TBTs (regulations 84 and 104) mandated rigorous testing and certification, entailing substantially higher compliance intensity. If the upgrading effects, that we document, were purely driven by reduced import competition—a channel common to all TBTs—we would expect β_1 to be positive and significant. The fact that retail TBT effects are small and insignificant across our outcomes, while the automotive interaction (β_2) is large and significant, provides direct evidence that compliance requirements—not import protection per se—drive the firm level changes.¹⁷

Theoretically, the sign of β_2 is ambiguous. On the one hand, compliance standards imposed by automotive TBTs may induce firms to invest in R&D, upgrade their inputs, and/or develop new products, thereby generating a positive effect on firm performance. On the other hand, reduced import competition could increase the market power of incumbent firms, weakening their incentive to innovate, and thus depressing long-run competitiveness. The net effect is an empirical question that depends on whether the *compliance* channel (upgrading to meet regulatory standards) or the *protection* channel (reduced competitive pressure) dominates.

An important concern is whether there were any other policies that coincided with the TBTs. And if such is the case, that could confound our estimates. In order to control for any aggregate macroeconomic shocks that are common to firms in both treated and control groups, we include year fixed effects γ_t .¹⁸ We also control for firm fixed effects, α_i , to control for all possible time-invariant firm characteristics. We cluster standard errors at the industry level (4-digit NIC), the level at which treatment is assigned.

However, a few other concerns still remain: (i) effect of concurrent policies which

¹⁶Since $Automotive_{ij}$ is a strict subset of $Treated_j$ —all automotive firms are treated, though not all treated firms are automotive—the interaction $Post_t \times Automotive_{ij}$ is algebraically identical to $Treated_{jt} \times Automotive_{ij}$ and cannot be separately included.

¹⁷As a robustness check, we re-estimate all of our main specifications by dropping retail-TBT firms and using only the untreated firms as the control group. The results are virtually unchanged, confirming that the composition of the control group does not drive our findings.

¹⁸During our sample period, India was simultaneously liberalizing tariffs across manufacturing industries. To ensure that our results are not confounded by changes in trade protection through the tariff channel, we re-estimate all our main specifications controlling for input and output tariffs at the 4-digit industry level. Our baseline estimates remain virtually unchanged. More on this later.

may have affected the automotive sector differentially; (ii) selection effect; and (iii) endogeneity of TBTs. We discuss each of them below:

3.1 Concurrent Policies

India implemented the 2002 Auto Policy which completed the FDI liberalization process that began in 1991 (when direct FDI in the automotive and few other sectors was capped at 51%), by raising the limit to 100% automatic approval. However, two features of our results confirm that FDI liberalization is not one of the primary drivers of our findings: (i) first, effects (at the firm level) are concentrated among only the *domestic* and not *foreign* firms. If FDI liberalization is the dominant channel, we would have observed the effects to be driven by the foreign firms, which are the direct beneficiaries of relaxed ownership caps; (ii) second, our results while exploring the decomposition effect of Regulation 84 vs. 104 show that upgrading is driven entirely by the ARAI certification requirement (Regulation 104), with no additional effect from the import ban (Regulation 84). FDI liberalization was a sector-wide, and not specific to certain firms, policy that should have affected all automotive firms uniformly, yet the upgrading effects are specific to the compliance-based regulation—a pattern consistent with the TBT channel but not with a generic FDI story.¹⁹

3.2 Selection Effect

Our firm level panel may be subject to survivorship bias if the TBTs induced selective exit among firms. In order to check whether such is the case or not, we provide direct evidence that mitigates this concern in **Table C.5**. Automotive TBTs *increased* the probability that firms would remain active and reduced amalgamation, thereby providing no evidence on selection effect driving our results.

3.3 Endogeneity of TBTs

One issue that may plague our estimates is if these trade-restrictive regulations were driven by market power of firms from a protectionist and discriminatory motive standpoint rather than the objective(s) of broader regulatory improvements which are unrelated to domestic industry performance. In order to investigate this, we analyze the relationship between past industry performance and the likelihood of imposing a

¹⁹Nonetheless, we note that the observed firm level effects may partly reflect the combined effect of the TBTs and complementary policy initiatives, and that FDI liberalization may have indirectly affected domestic firms through anticipated competitive pressure or technology spillovers.

restrictive TBT. **Table 2** presents the required results.²⁰

We start by examining the role of import penetration by considering industry level import flows in columns (1) to (3). For these columns, we use product level import data from the UN Comtrade database, accessed via the World Bank’s World Integrated Trade Solution (WITS). This dataset provides comprehensive coverage of merchandise trade at the HS 4-digit level across virtually all traded goods (1,241 sectors in our sample). The results show a clear and consistent pattern: higher import penetration is associated with a greater likelihood of a restrictive TBT.

In contrast, columns (4) – (6) use firm level data from the PROWESS aggregated to the industry level to examine the relationship between an industry performance (as measured by markups) and the incidence of a TBT. We source the markup data from [Singh & Chanda \(2021\)](#).²¹ Our results show no statistically significant relationships with industry level performance or monopoly power and TBT incidence. This suggests that the imposition of trade-restrictive regulations may not be driven by lobbying efforts from the most powerful firms in these industries.²²

Taken together, these results highlight two key insights: (i) first, import penetration appears to play a significant role in determining which industries are targeted by trade-restrictive regulations; and (ii) second, little evidence to suggest that these regulatory changes are the result of lobbying efforts or are motivated by protectionist interests tied to industry performance. Therefore, we can treat these regulations as representing broader regulatory efforts unrelated to the direct influence of firms’ market power.

We structure the following analysis around three main steps. First, we examine whether TBTs reduced import flows and reflected any changes in firm capabilities, focusing on investment in R&D and technology transfer. Second, we move to granular margins by utilizing firm-input and firm-product level data to examine how firms adjusted their input mix and product portfolios. Finally, we investigate whether these firm level adjustments translated into broader market outcomes, including domestic market share, export development, and India’s revealed comparative advantage in the automotive sector.

²⁰Note that the difference in the number of observations across columns reflects differences in data sources and levels of aggregation.

²¹The data on markups is restricted in coverage to 64-65 industries at the 4-digit level with sufficient cost and revenue information.

²²A remaining concern is that average markup may mask the influence of top producers: if a few dominant firms with exceptionally high markups lobbied for TBT protection, the industry mean could still appear unrelated to TBT assignment. To address this, **Table C.6** extends the markup analysis by replacing the mean with 75th- and 90th-percentile markups, computed from the firm-product distribution within each NIC 4-digit industry. All nine specifications—testing lagged levels, lagged growth, and pre-period averages for mean, 75th-, and 90th-percentile markups—yield statistically insignificant coefficients. Notably, the upper-tail coefficients are uniformly *negative*, suggesting that industries with the highest-markup producers were, if anything, *less* likely to receive TBTs. This rules out the possibility that powerful top firms drove the regulation to protect their market position.

4 First-order Effects: Import Flows

Before estimating the effect on firm level outcomes, such as technology adoption expenses, input usage, etc., we first check what happened to import flows as a result of the TBTs. The TBTs (automotive) banned the import of second-hand vehicles older than three years and imposed a special certification for the younger ones. Therefore, all these restrictions, in theory, should reduce import flows, at least in the short run. In order to check this hypothesis, we download data from WITS (World Integrated Trade Solutions) at the HS 4-digit level and match it with our TBT dataset. We present our results in **Table 3**.

Columns (1) – (3) run OLS while, (4) – (6) use PPML method. Column (1) aggregates imports across all countries by HS 4-digit level and regresses on our variable of interest, $Treated_{jt}$, controlling for industry, year fixed effects and interactions of industry-year trends. We find strong negative effects. The imposition of the TBTs led to about 37% drop in imports of those product categories on which they were imposed. Column (2) additionally introduces the interaction of $Treated_{jt}$ with $Automotive_{ij}$ to estimate the differential effect of the automotive TBT. The interaction term is negative, but imprecisely estimated. However, when we calculate the joint effect for the automotive TBTs, we find that to be large and significant. Column (3) repeats column (2) but using import flows at the product-country-year level and not aggregating imports by HS 4-digit level i.e., at the product-year level. We continue to find negative effect of the TBTs on import flows, but now with a significant differential effect for the automotive TBT as well. Columns (4) – (6) re-run columns (1) – (3), but using PPML. The findings are qualitatively similar. Automotive regulations or TBTs have a strong negative impact on import flows.^{23, 24}

²³**Figure D.1** plots the event study specification by interacting year fixed effects with $Automotive_{ij}$. We find evidence of strong negative effects after the imposition of the TBTs.

²⁴**Table C.7** explores the effect of TBT on import flows by varying country groups, and our thought process here is to check whether there is any heterogeneous effect across different country groups or not. We use three different country groups – (i) high-income, (ii) OECD, and (iii) complainant and its interaction with $Treated_{jt}$ as variables of interest. We find significant negative effects on import flows across all these three country groups.

5 Effect on Production Capabilities: Technology Adoption, Inputs, and Products

5.1 Technology Adoption

A key implication of the automotive regulations or TBTs was that they introduced several mandatory tests for all vehicles and their components, and this required a certification from ARAI. Also, at that point in time ARAI did not have testing facilities for large machines, therefore firms had to set up their own facilities or use those of others who had a similar set up. All these provisions of the automotive regulations can significantly alter the technology adoption expenses of a firm, in particular both R&D and technology transfer. PROWESS is unique in the sense that it not only gives details on expenses made due to R&D and technology transfer, but also on its sub-components, such as R&D capital, R&D current, expenses incurred due to technology transfer from domestic sources and similarly from foreign sources. We use all these variables as our outcomes of interest and report the results in **Table 4**.²⁵

We start by looking at the total technology adoption expenses, which is the sum of R&D expenses and expenses on technology transfer, in column (1) of **Table 4**. Our estimate shows that regulations 84 and 104 or the automotive TBTs led to a 38% increase in total technology adoption expenses for a firm affected by those TBTs compared to others. Next, we divide the total technology adoption expenses into R&D (columns (2) – (4)) and technology transfer (columns (5) – (7)). The automotive regulations specify that firms need to carry out tests for the engines of new vehicles, as well as the inputs or components used in making a new vehicle. These provisions can significantly affect both capital and current, more so the former, R&D expenses as some firms may have needed to set up the testing facilities and/or carry out new R&D for using new inputs to produce the new products. The automotive TBTs led to an increase in both types of R&D with about 10% higher for capital R&D expenses (22%) than current (20%). On the other hand, we find evidence of increase in technology transfer expenses for both

²⁵Please note that we use $\ln(1 + x)$ transformation for R&D and technology transfer expenditures, which retains firms reporting zero expenses in the estimation sample. This ensures that our estimates capture both the intensive margin (how much more existing innovators spend) and the extensive margin (whether non-innovating firms begin to invest). Many Indian manufacturing firms, particularly smaller ones do not report any formal R&D spending, so preserving these observations is important to avoid selection bias.

domestic and foreign sources, with about 50% more from domestic sources.^{26,27,28}

Event-Study Plots: A causal interpretation of our reduced form coefficients is subject to the assumption that technology adoption (for that matter both R&D and technology transfer) expenses for firms in the automotive TBTs industries and others would have evolved comparably in the absence of the regulations. Although the counterfactual cannot be directly tested, we exploit data prior to the policy change to estimate event-study specifications. In particular, we interact year fixed effects for three years prior to the implementation of the TBTs, and also seven years after, with $Automotive_{ij}$.²⁹ We use the following regression equation:

$$\ln(x_{ijt}) = (\lambda_t \times Automotive_{ij}) + \eta_i + \phi_t + \epsilon_{ijt} \quad (2)$$

Figure 3 plots the event study graphs by estimation Equation (2) for technology adoption (**Panel A**), R&D (**Panel B**), and technology transfer (**Panel C**) expenses. Our plots show little to no evidence of pre-trends in case of all our three outcome variables of interest for firms in automotive TBTs industries compared to others with strong and significant differences afterward, i.e., post the implementation of the TBTs. **Figures D.2, D.3, and D.4** plots the event study graphs for dividing R&D expenses into capital and current, technology transfer expenses into domestic and foreign, and controlling for industry-year time trends, respectively. We find all our plots to be consistent with the diff-in-diff results.³⁰

²⁶The TBTs by design are bound to reduce competition in the domestic market by restricting import flows (as shown before). This can increase the market power of firms, especially domestic firms, which are focused on the domestic market. An increase in monopoly power can have either adverse or no incentive for firms to upgrade their technology adoption. However, this would depend on the nature of the TBT, specifically whether it applies similarly to domestic firms or not. We use a few of the key firm level characteristics, such as ownership, export status, pre-policy technology capacity, to check which types of firms drive the overall effect in **Table C.8**. Our results show that the effect is driven by domestically-owned firms, which are exporters, and belong to above median of the technology capacity distribution.

²⁷Another key question that immediately comes up is whether we are using the correct comparison or control group. Our current control group comprises of firms which are also treated with retail-TBTs along with firms which are completely untreated. **Table C.9** drops the firms which were treated with retail-TBTs and use only untreated firms as the control group. We find very similar results, even quantitatively.

²⁸**Table C.10** controls for input and output tariffs at the 4-digit NIC level (following [Chakraborty & Raveh \(2018\)](#)) to check if our results are driven by the correlation between the imposition of TBTs and the simultaneous trade reform. We find no such effect – our benchmark results hold.

²⁹ $Treated_{ij}$ continues to assume a value 1 if a firm i belongs to an industry for which the automotive TBTs was implemented.

³⁰We note that the event-study coefficients for capital R&D (**Figure D.2, Panel A**) merit particular attention. One might expect capital investments—such as setting up testing facilities—to respond with a lag. However, the automotive TBTs were notified to the WTO and published in the Gazette of India in early 2001, and the underlying regulatory process began before formal implementation (see **Appendix A**). Firms that anticipated the regulatory change may have begun adjusting their investment plans in advance. Moreover, the coefficients build progressively over the post-treatment period rather than appearing as a discrete jump, consistent with a gradual investment response to the new compliance re-

5.1.1 Who and What is Driving the Effect?

The reduced-form evidence that we find with respect to technology adoption in the previous section establishes that automotive TBTs is associated with substantial upgrading, but it does not by itself identify the operative channel. Two concerns remain. First, automotive regulations are policy bundles: compliance requirements and import restrictions are introduced together, so average treatment effects cannot separately attribute the response to certification-induced capability investment versus pure protection. Second, the baseline comparison uses non-automotive controls (retail-TBT and untreated firms) that may differ in latent sectoral trends, technology opportunities, and supply-chain structure, even with fixed effects and event-study support.

To pin down the mechanism, we first separate the compliance component (Reg. 104) of the automotive policy package from the pure import-restriction (Reg. 84) component. In other words, we estimate the effect separately for Reg. 84 and 104. This is important to understand whether a simple import ban can act as a *de facto* industrial policy or a combination of compliance requirements is necessary for the observed effect. We present our results in **Table 5**. Our estimates show that the overall effect for the automotive sector in terms of adoption of higher R&D (more capital than current) and technology transfer (more domestic than foreign) is overwhelmingly driven by Reg. 104. This result also shows that how a compliance requirement may turn into a targeted industrial policy.³¹

Next, we keep the sample restricted to only automotive firms and interact five key pre-treatment characteristics – firm undertaking R&D or not, large firm (if a firm’s assets is above the median of the asset distribution of the corresponding sector), exporter, importer, and maker of vehicle parts – to test whether the mechanism operates more strongly among firms with specific traits in **Panel A** of **Table 6**. We find that firms which are import-dependent exhibit large increase in technology adoption, R&D, and technology transfer.³² **Panel B** further divides import dependency into four different categories – importer of capital goods, importer of raw materials, importer of stores & spares, and importer of finished goods – and check which type of importer reacts the highest with respect to technology adoption measures. We find that importer of capital goods had the highest effect followed by importer of raw materials. It is interesting to

quirements.

³¹**Table C.11** further establishes our argument. It presents a within-automotive test that restricts the sample to automotive industries only and estimates the effect for the additional Reg. 84 relative to the common Reg. 104 compliance regime. The estimated additional Reg. 84 coefficients are not statistically significant across technology adoption, R&D, and technology transfer reinforcing that the compliance channel (Reg. 104) drives the adjustment.

³²**Table C.12** runs five different sets of regressions using these characteristics and find import dependence to be the most important characteristic which explains the observed increase in technology adoption.

note here that both these types of firms have similar effect for R&D expenses, but the former have significantly higher effect (around 40%) for technology transfer. Together, these results point to supply-chain exposure, rather than broad firm type, as the main amplifier of the treatment effect.

Finally, we use the capability-reallocation test in **Panel C** to explain why import-dependent firms responded more strongly. If compliance requirements are binded more tightly for firms that previously relied on imported capital goods and components, those firms should adjust by replacing external imported capability inputs with domestic and in-house capability accumulation. This is exactly what we find in **Panel C**: more exposed firms reduce imported capital-goods reliance while increasing domestic technology transfer and R&D capital. The reallocation pattern provides micro level adjustment margin behind the larger treatment effects among import-dependent firms.³³ Taken together, the evidence supports a “forced quality upgrading” interpretation in which compliance-induced capability investment is the main transmission channel.³⁴

5.2 Input Mix

A key follow-up question that arises given the observed increase in technology adoption is: what happened to the input composition of those firms? The regulations (no. 84 and 104) made testing mandatory for new vehicles and their components to get a required certification from ARAI. Therefore, the requirement for testing and certification to sell a final product can have strong spillover effects in terms of inputs or components used by firms belonging to that sector. In other words, the regulations can significantly influence automotive firms to use higher quality inputs in their production process leading to an input quality-upgrading mechanism.

PROWESS provides detailed data on the total expenses paid for inputs used, name of each of the input used, the value and quantity of each of the input used by a firm towards production of its final products. We utilize such information and examine the effect on the automotive TBTs on the number of inputs used, value of each input, and average input expenses and report our results in **Table 7**.

We start by looking at the total number of inputs used by a firm in its production as the outcome variable of interest in **Panel A** of **Table 7**. We count the names of the inputs used by each firm for every year and estimate the total number of inputs used by

³³We also check for the effect on R&D current and foreign technology transfer expenses in **Table C.13**. We find similar effects.

³⁴**Table C.14** reports the same test for importers dependent on raw materials, stores and spares, and finished goods. Only firms who had pre-policy dependence on raw materials is associated with higher R&D capital and domestic technology transfer with no to weaker effects for importers of stores and spares and finished goods along these margins.

an individual firm in column (1). We find that the TBTs led to a significant increase, 6%, in the number of inputs used by a firm in the automotive industries compared to others.

Columns (2) and (3) use information on the value of each input used by a firm by utilizing information at the firm-input level. Our idea is to check whether the TBTs led firms to use more high-quality inputs in their production function given that it is mandatory to procure a certificate from ARAI for parts and components of the vehicles or their final products as well. For these regressions, we replace firm with firm-input fixed effects as our data varies now at the firm-input-year level. Additionally, we control for the price of inputs (total value/total quantity) in column (3). We find that value of an input used by a firm belonging to the automotive industries went up by 23–26%. Lastly, we check for average input expense as our outcome variable in columns (4). We sum expenses across all inputs and divide it by the number of inputs used to estimate the average input value. We find a very similar result – the value of an average input also went up by 28%.^{35,36}

We plot the event-study graphs for the average input expenses and input value of each input used by a firm in **Figure 4**. **Figure D.5** repeats the exercise for only the domestic firms. Our plots show that there were no differential trends in terms of inputs used before the implementation of the TBTs for the automotive firms and it started increasing afterward and these aggregate changes are completely driven by domestic firms.

Lastly, in **Panel B** of **Table 7** we separately estimate the effect for Reg. 84 vs. 104. We find that the overall positive effects on input mix is driven by Reg. 104. Interestingly, Reg. 84, which is a pure import ban, has an opposite effect on the value of inputs used. This result further supports our previous claim that an import ban along with a compliance requirement acted as a *de facto* industrial policy for the automotive sector which lead to a upgrading of the production technology of that sector.

³⁵**Table C.15** checks for heterogeneity in results by dividing firms by ownership, exporters vs. non-exporters, and below and above median technological capacity. Likewise our results on technology adoption, our findings in case of inputs used are overwhelmingly driven by domestic firms which are non-exporters. Interestingly, while dividing the sample of firms in terms of their technological capacity, our estimates show that firms below the median of the technological capacity distribution also started using more expensive inputs (28%). This shows that the effect of the TBTs in terms of the input quality upgrading is not concentrated in a certain section of firms, but is pervasive across the technological distribution of firms.

³⁶**Table C.16** controls for input and output tariffs at the 4-digit NIC level and our findings on input scope, value of inputs used, and average input expenses turn out to be robust to the simultaneous drop in tariffs.

5.3 Product Portfolio

Did this change in input composition and/or use of higher value inputs by automotive firms have any implication for their product portfolio? In other words, whether the TBTs led to introduction of new products by the automotive firms or they continued to produce old products with better inputs? Additionally, what happened to the price, markups, and marginal costs of those firms? These questions are very important in order to understand how the dynamics of the sector changed as a result of these protectionist policies or TBTs imposed by India. We use detailed firm-product level data from PROWESS to answer these questions. Results are reported in **Table 8**. Unless otherwise specified, we use firm-product fixed effects for all our regressions.

We start our analysis on product portfolio of firms by summing all the products produced by a firm to estimate the product scope of a firm in column (1). Our data now varies at the firm-year level, so we use firm fixed effects rather than firm-product fixed effects. Our results show no significant changes in the product scope of a firm. Column (2) substitutes product scope of a firm by product entry. It takes a value 1 if a new product has been introduced or produced by a firm in the list of all its products produced given that it has not been produced before in our sample time frame. We continue to find little to no evidence of any effect of the TBTs on new products manufactured by automotive firms. However, the joint effect shows us a negative effect, suggesting that TBTs reduced the probability of introducing a new product.

Column (3) replaces product entry with exit. We find that the effect of the TBTs on product exit to be significant and negative. The probability of dropping a product by the automotive firms went down by 3%. This suggests an interesting point: automotive firms, as a result of the protectionist policies, consolidate their existing product lines with new high-quality inputs rather than diversifying or dropping existing ones and replace them with new.^{37,38}

This brings us to our next question on what happened to the price, markups, and marginal costs of those firms. Price can increase due to two reasons: (a) expenses of firms have increased due to R&D, technology transfer, and higher value of inputs, and firms would like to pass-through that increase in costs to the consumers; and/or (b) rise in protectionism increased the market power of firms and this led to an increase in prices. PROWESS, the dataset we use, is unique in the sense that it provides value of sales and quantity sold for each of the products produced by a firm. We follow [De Loecker et al. \(2016\)](#) to calculate the price, markups, and marginal costs at the

³⁷We check for the robustness of the results by controlling for input and output tariffs at the 4-digit NIC level in **Table C.17**. We do not find any change.

³⁸**Figure D.6** shows that TBTs also had a positive effect on product sales and quantity for the automotive sector as well with no trends in the differences between automotive firms and others.

firm-product level in columns (4) – (6), respectively.

We find that the price of the products produced by a firm increased by 19%.³⁹ Note that this number is close to what we find in terms of the increase in average value of inputs used by firms. We find that this increase in price is driven by an increase in markups (34%) and not costs. Interestingly, we find a drop in marginal cost of producing a product, but the standard error is imprecisely estimated. Our findings are similar to what [Singh & Chanda \(2021\)](#) finds.⁴⁰

An important caveat concerns the interpretation of the markup result. Combined with the significant price increase, this pattern suggests that markup gains are primarily driven by the exercise of greater market power under reduced foreign competition, rather than by efficiency improvements with lower production costs. In other words, while the technology adoption and input upgrading results point to genuine capability building on the production side, the markup evidence is more consistent with a standard competition effect. We therefore interpret firm level upgrading as operating primarily through the production process–technology adoption, input quality, and product consolidation–rather than through improved cost efficiency.

Overall, our analysis shows that firms did not respond to the TBTs by expanding their product portfolios through the introduction of new lines. Instead, they deepened specialization with upgrading of input quality and increasing their market power. This pattern indicates a strategic focus on strengthening established market positions under the new regulatory environment.

6 What happened to Sales?

As shown before, the TBTs had trade distortionary effects in terms of negative effects on import flows. This can have an opposite effect on domestic sales for domestic automotive firms and increase the market share for those firms in the home market. We now check whether such is the case or not. In addition, we also examine whether the changes in the production process led some of the domestic automotive non-exporters to become exporters. **Table 9** presents result from such exercises.

Columns (1) – (3) use domestic sales of a firm as the outcome variable of interest. We estimate the results for all firms, domestically-owned firms, and firms which are non-exporters. We find that the TBTs led to an increase in domestic sales by about 30%.

³⁹Note that the number of observations drops in the case of estimating the price as we only focus on those firms for which there was consistent reporting of the data on sales value and quantity sold.

⁴⁰All of these results on prices and markups continue to be driven by domestic firms (Results available on request). We also check these results at firm rather than at the firm-product level. We find similar results in terms of increase in the markups of the automotive firms (Results available on request).

This is highly significant for both domestically-owned and non-exporters, especially for the latter group of firms. Domestic sales for non-exporters increased by about 50% whereas for domestically-owned firms this is about half of non-exporters. **Figure D.7** presents the coefficient plot of domestic sales for non-exporters. It shows that while the trend is negative (not significant though) before the imposition of TBT, it turned in the opposite direction afterward.⁴¹

Next, using exports as the outcome variable of interest in columns (4) – (6), we find that exports also increased by 22% for firms belonging to the automotive sector, but interestingly this increased more for firms which were non-exporters before the imposition of the TBT. In other words, TBTs led non-exporting automotive firms to become exporters. To confirm this, we ran another regression in column (7) using the probability of exporting after 2001 as the outcome variable. In particular, the variable takes a value 1 if a firm registers a positive amount in exports after 2001 given that it has never exported before 2001. We find that the probability of exporting increased by 6 percentage points at the mean. To put this in context, the baseline probability of a non-exporting automotive firm transitioning to exporting in our pre-treatment period is low, making this a substantively large effect on the extensive margin of trade.⁴² **Figure 5** carries out the event-study plots of domestic sales and exports of automotive firms as compared to others. We find a very similar trend as before. Differences in domestic sales and exports for automotive firms started to become significant after 2001 with no discernible difference before.⁴³

Given this result on increase in both intensive and extensive margin of exports, we ask the following question: did all the changes lead to an increase in the share of

⁴¹A natural follow-up question arises: did this increase in domestic sales result in an increase in domestic market share for domestic producers? An increase in domestic sales may not directly result in an increase in the market share of the non-exporting or domestically-oriented firms. It depends on how much domestic sales have increased or imports dropped. One way to check this would be to calculate the share of domestic sales as a share of total sales in the Indian domestic market for the automotive industry. **Figure D.8** estimates the share of domestic sales to total output (Imports + Domestic Sales) for the automotive sector (Panel A) and how much of this share is driven by domestically-owned firms (Panel B). To carry out this exercise, we first aggregated domestic sales across all firms in the automotive sector to estimate the total domestic sales by firms in the domestic market. Secondly, we downloaded total import data for the automotive sector from the WITS World Bank and added them with the domestic sales to estimate the total amount of sales for the automotive sector in the Indian domestic market. We then calculated the share of domestic output in total sales of automotive sector firms from 1996 to 2005. We find that the share of domestic sales jumped from about 70% to more than 80% in 2001, as when the TBTs were implemented, and then hovered around 80% for the following years. This shows that TBTs not only changed the automotive firm production process using better quality inputs and increased market power through increase in markups, but also led to an increase in the domestic market share of these firms. This may also explain why the markups increased for those firms.

⁴²The exact baseline transition rate can be computed from the pre-2001 sample. The 6 percentage point increase should be interpreted relative to this baseline.

⁴³**Tables C.18, C.19, and C.20** decompose the effect by estimating the Reg. 84 and 104 separately, use only untreated firms as the control group, and control for tariffs (input and output), respectively. Our benchmark finding on sales and exports continue to remain the same.

exports of the automotive sector? Or did the ratio of the exports of the automotive sector increase compared to other industries? Or did the changes in the production process in terms of adopting higher technology, using high-quality inputs, producing high-value products lead to an increase in India's revealed comparative advantage in that sector? Using data from WITS-World Bank at the HS 4-digit industry level, we seek answers to these questions.

6.1 Revealed Comparative Advantage

Figure D.9 plots the exports of the automotive sector in three different ways. **Panel A** simply calculates the share of automotive exports in total exports of India. It shows that the ratio was about 15% in 2001 and it increased to 35% by 2005, more than double within a span of 5 years. **Panel B** compares automotive exports with exports of all other industries in terms of calculating a ratio of automotive exports to all other exports by India. This ratio also increased from about 2% in 2001 to 3.5% in 2005. Lastly, following [Lane \(2025\)](#), we calculate the revealed comparative advantage (RCA) of Indian automotive exports. We find that the revealed comparative advantage also increased from 0.225 in 2001 to 0.35 in 2005. And, for all these indicators the ratio was either declining or stagnant from 1996 onward till 2001–02.⁴⁴ We also plot event study graphs using India's export shares of the automotive sector as the outcome variable using both OLS and PPML in **Figure D.11**. We find for both the cases export shares of automotive sector starts to be significantly different from zero after 2001, especially for the PPML estimation.

Cross-Country Comparison. A natural question is whether this RCA gain of India simply reflects a global automotive trend – for instance, a worldwide expansion in automotive trade that would have benefited India regardless of the TBTs. To address this, we compute the RCA index for all countries with available COMTRADE export data over 1997–2005 (matching the firm level analysis window) and estimate a difference-in-differences regression of the form:

$$RCA_{ct} = \alpha_c + \alpha_t + \beta(\text{India}_c \times \text{Post}_t) + \varepsilon_{ct} \quad (3)$$

where country fixed effects absorb level differences and year fixed effects absorb any common global trend in automotive RCA. **Table 10** reports the results. Across all countries, India's automotive RCA increased significantly after 2001, by 12.2%. Impor-

⁴⁴Additionally, **Figure D.10** estimates India's RCA index of automotive exports with respect to three different country groups – advanced, low-middle income, and least developed. The figures reveal that India's RCA index increased, especially in comparison to advanced and not to least developed economies. This shows that quality upgrading of the sector, due to the TBTs, led to significant gains from exports as well.

tantly, when dividing the comparison to developing and advanced economies, the effect is shown to be driven the latter set of countries, indicating that India's RCA gain was not part of a generic emerging-market industrialization trend. **Figure 6** presents the event-study version of this regression. The pre-2001 coefficients are negative, consistent with India's declining automotive competitiveness prior to the TBTs—precisely the deterioration that motivated the policy intervention. The sharp reversal after 2001, coinciding with the TBT introduction, is consistent with the firm level upgrading results documented above translating into aggregate export competitiveness.

6.2 Welfare Implications: A Back-of-Envelope Assessment

An immediate question that crops up is whether the producer gains documented above justified the costs borne by consumers through higher prices or not. We provide a partial-equilibrium calculation using our difference-in-differences estimates—which isolate the causal effect of the TBTs—and pre-treatment (1997–2000) averages from PROWESS and COMTRADE. **Table C.21** reports the full decomposition.

On the consumer side, using pre-treatment annual averages of domestic sales of automotive firms, which is INR 464 billion (USD 11.1 billion), and the estimated 19% price increase (**Table 8**, column (6)) implies an annual consumer cost of approximately INR 88 billion (USD 2.1 billion). On the producer side, the 33% increase in domestic sales and 24% increase in exports, evaluated at a pre-treatment profit margin of 20%, yield annual producer surplus gains of approximately INR 32 billion (USD 758 million) – roughly one-third of the consumer cost ($PS/CS \approx 0.36$). This ratio rises to 0.54 under a 30% margin but remains below unity under any plausible parameterization. In purely static terms, consumer losses exceeded producer profit gains.

However, three dynamic considerations temper this conclusion. First, the TBTs induced approximately USD 50 million per year in additional technology adoption—a durable investment in capabilities, not a deadweight loss. The Regulation 84 vs. 104 decomposition confirms that these investments were driven by compliance requirements, not by the import restriction itself. Second, the export expansion and near-doubling of India's automotive RCA represent genuine new surplus from world markets, not a domestic transfer. Third, the 29% increase in domestic sales primarily reflects import substitution as imports fell by 30% (PPML estimate, **Table 3**, column (5)). This implies that domestic firms captured the vacuum in the market share, maintaining product availability albeit at higher prices. We note that these calculations are approximate: the consumer cost treats the full price increase as welfare loss, whereas part may reflect quality improvements (input unit prices did not rise), and the profit margin is

assumed.⁴⁵ Nevertheless, the magnitudes provide a quantitative benchmark for the welfare trade-off.

7 Conclusion

We estimate the firm level consequences of India’s 2001 automotive TBTs. Using detailed data on technology adoption, input use, and product portfolios, we show that a trade-distortionary policy produced *de facto* industrial policy effects: increased technology adoption, input quality upgrading, higher domestic sales and market share, and export expansion including the creation of new exporters and a near-doubling of India’s automotive RCA.

Firms concentrated on core capabilities—adopting more technology and upgrading inputs rather than diversifying their product lines—consistent with models where firms respond to changing market conditions by enhancing production complexity (Melitz & Redding, 2021). The decomposition of Regulation 84 vs. 104 reveals that it was the compliance requirement—not the import restriction—that drove upgrading, distinguishing this episode from classical infant-industry protection and aligning it more closely with the U.S. automotive experience under voluntary export restraints in the 1980s.

From a policy perspective, our partial equilibrium welfare assessment shows that static consumer costs were substantial, exceeding producer profit gains under any plausible margin assumption. However, the compliance-driven investment channel, export gains, and the durable nature of capability investments suggest a more favorable dynamic balance. More broadly, TBTs that impose compliance requirements can create incentives for capability investment that go beyond what a simple tariff would generate. Understanding when and how such regulatory instruments produce upgrading effects, rather than merely shielding incumbents, remains an important question for both trade and industrial policy.

⁴⁵The sensitivity analysis in **Table C.21** shows that the PS/CS ratio ranges from 0.18 (10% margin) to 0.54 (30% margin). We do not account for variety losses from reduced imports or general equilibrium adjustments.

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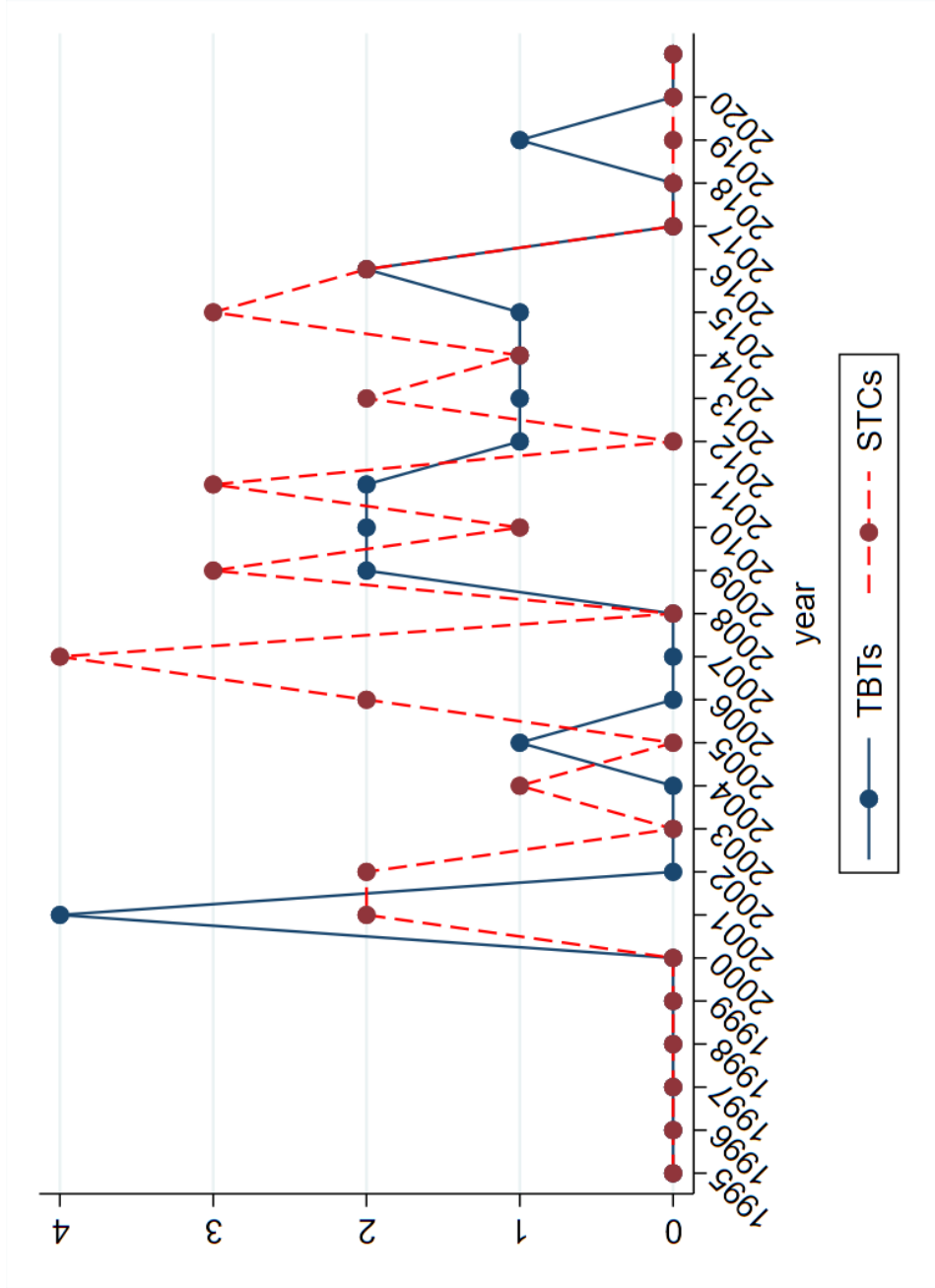


Figure 1: Frequency of Regulatory Measures Imposed by India on its Imports and Corresponding Trade Concerns Raised at WTO (1995–2020)

Notes: Figure represents the cumulative number of TBTs imposed by India on its imports yearly and STCs raised by other countries in the WTO. Authors' compilation from the STC dataset and Gazette of India.

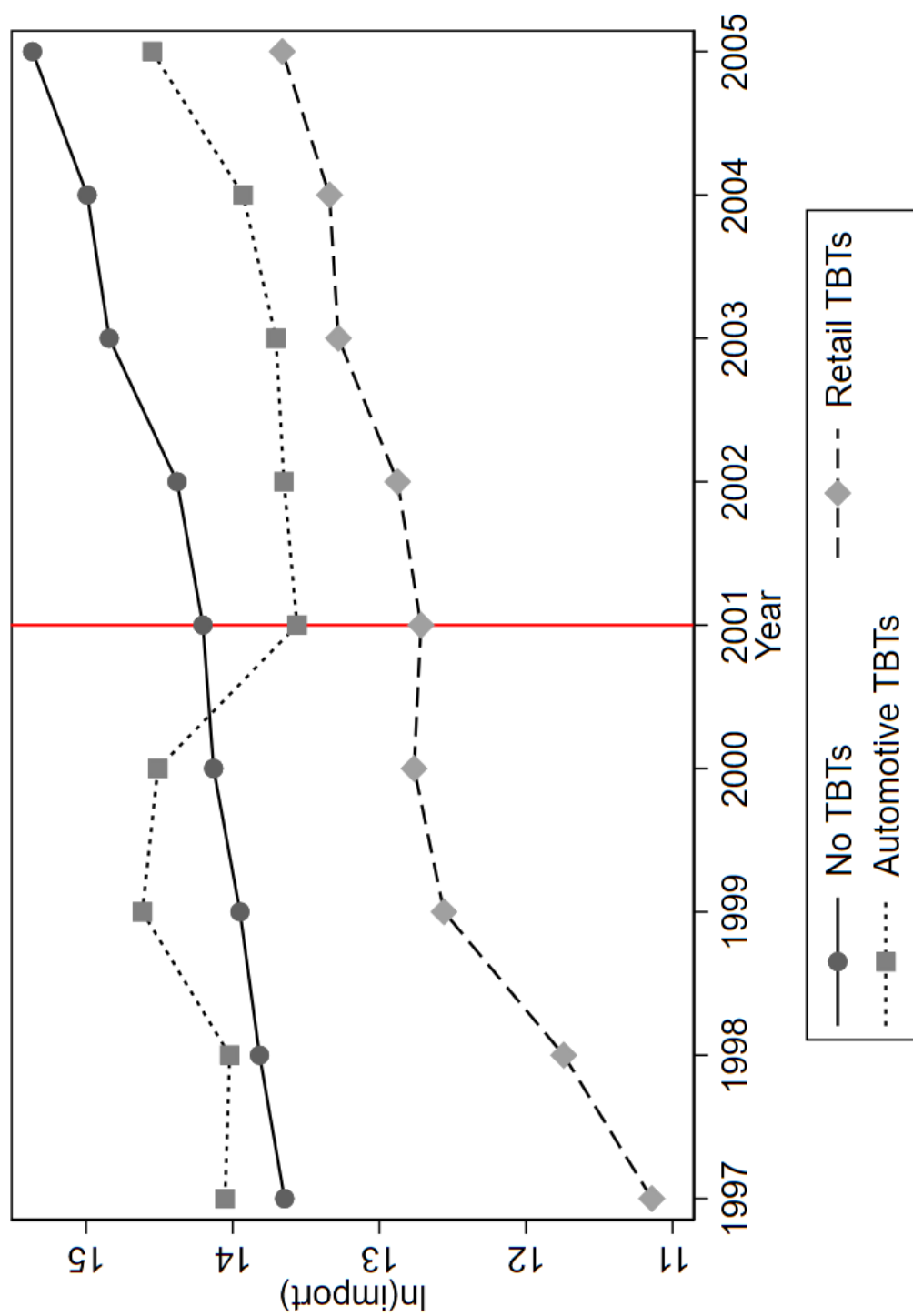


Figure 2: Average Industry Level Imports (Annual): by Concerned TBTs

Notes: Figure presents industry level (HS 4-digit) imports by different types of regulations.

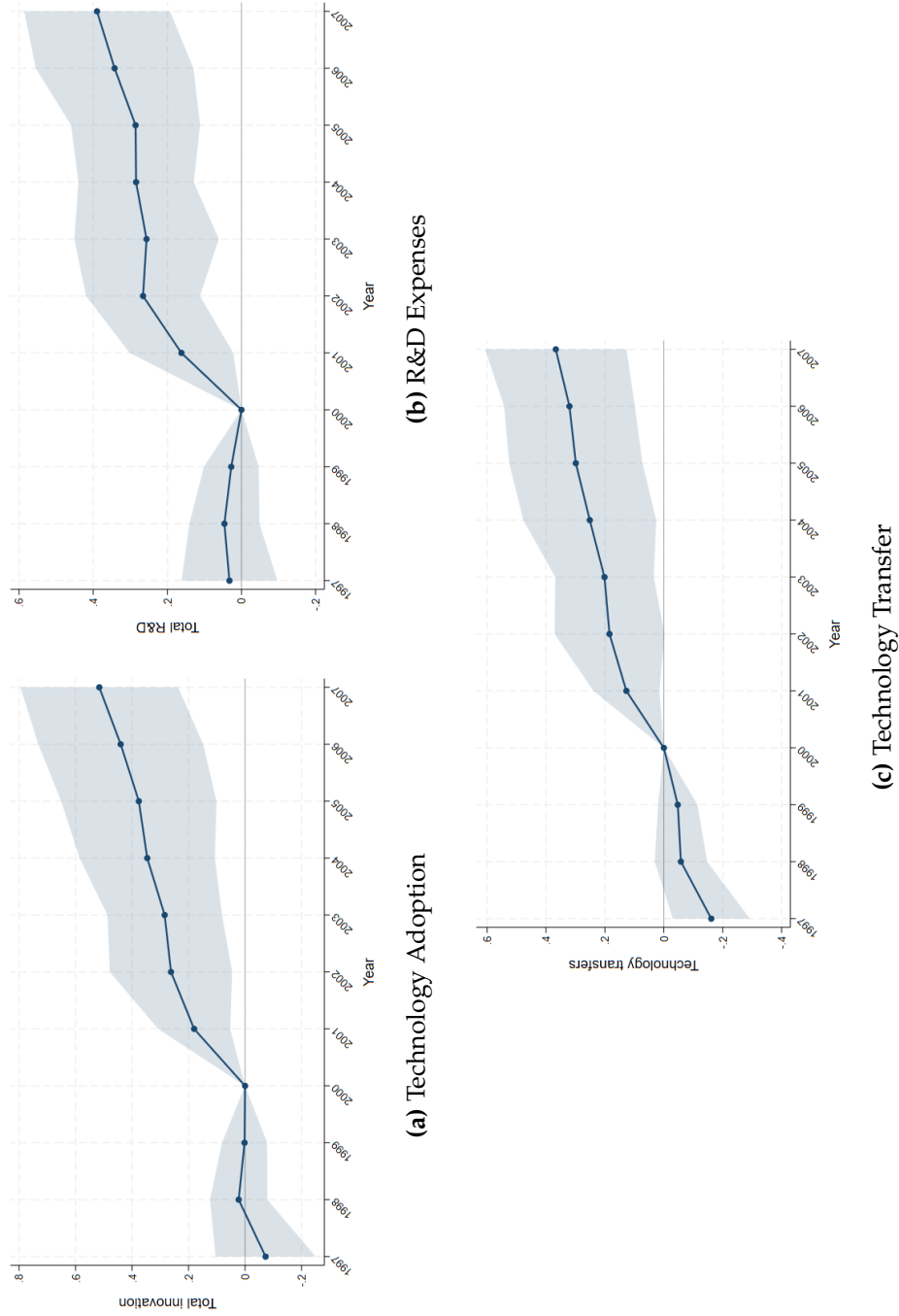
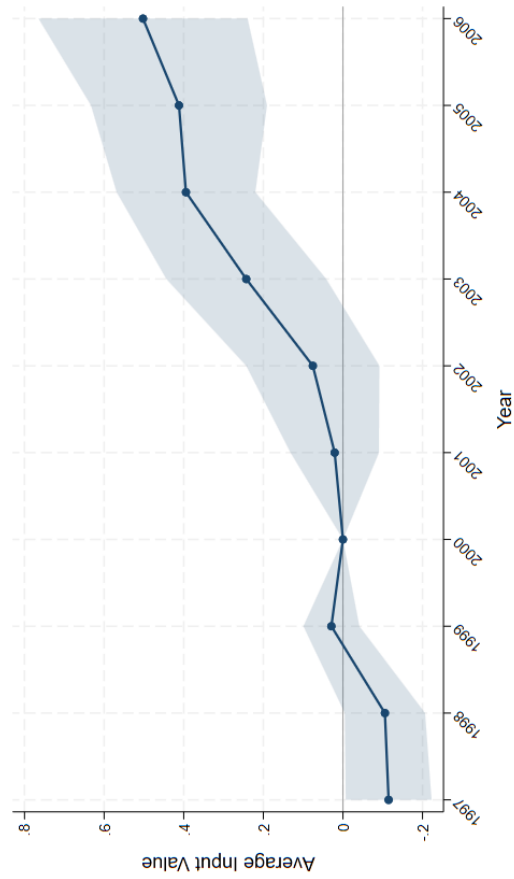
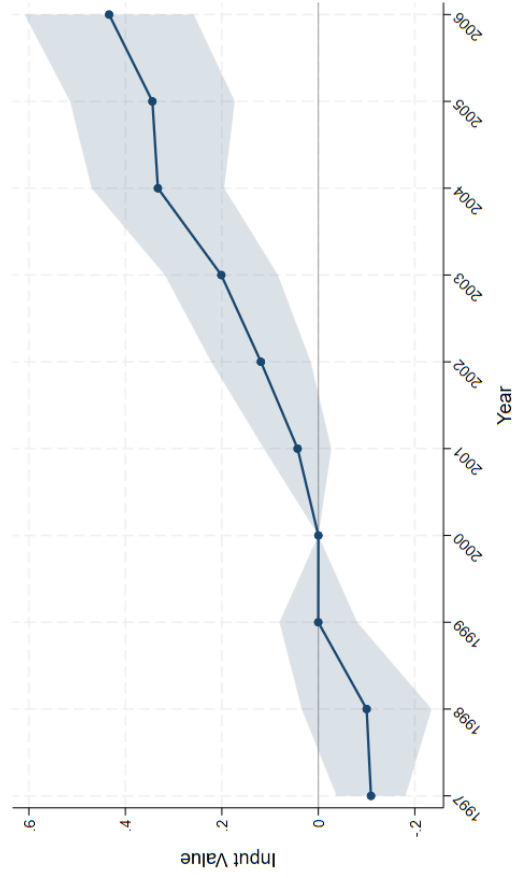


Figure 3: Automotive TBTs and Technology Adoption, R&D, and Technology Transfer Expenses

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the Total Technology Adoption expenses (Panel A), R&D expenses (Panel B), and Technology Transfer expenses (Panel C) of firms affected by automotive TBT with respect to all other firms. All other firms include firms (i) which are treated with the other regulation, i.e., retail TBT, and (ii) firms which are not affected by any TBT.



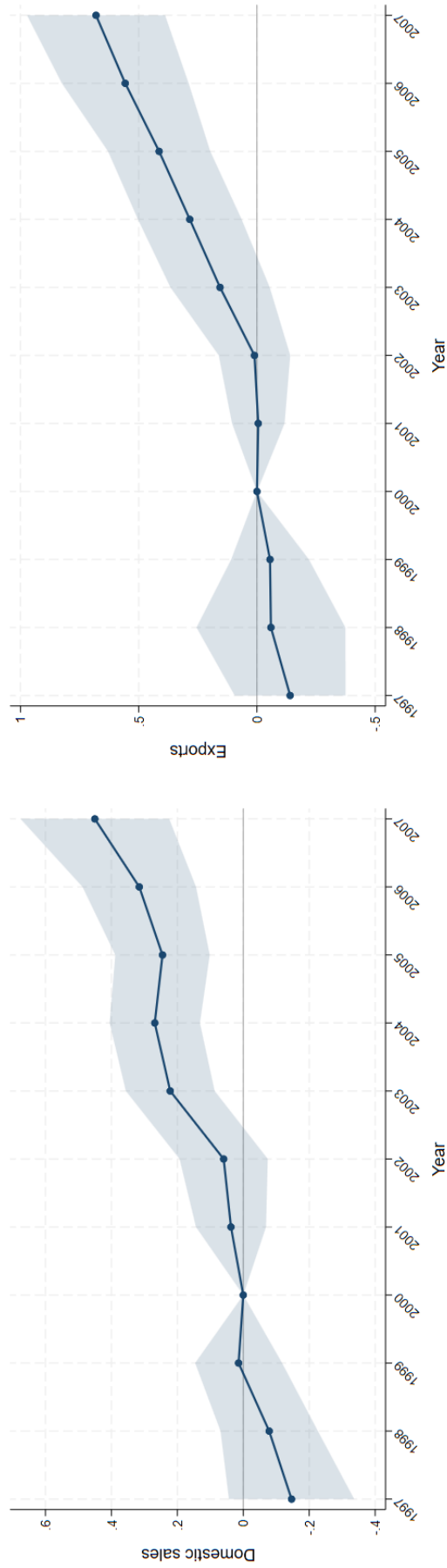
(a) Average Input Value



(b) Value per Input

Figure 4: Automotive TBTs and Effect on Input Value

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the firm average input value (Panel A), and value per input (Panel B) between the firms which are affected by the automotive TBTs (regulations no. 84 and 104) and all other firms. All other firms include (i) firms which are treated with the other regulations, i.e., retail TBTs, and (ii) firms which are not affected by any TBT.



(a) Domestic Sales

(b) Exports

Figure 5: Automotive TBTs and Effect on Domestic Sales and Exports

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the domestic sales (Panel A), and exports (Panel B) between the firms which are affected by the automotive TBTs (regulations no. 84 and 104) and all other firms. All other firms include (i) firms which are treated with the other regulations, i.e., retail TBTs, and (ii) firms which are not affected by any TBT.

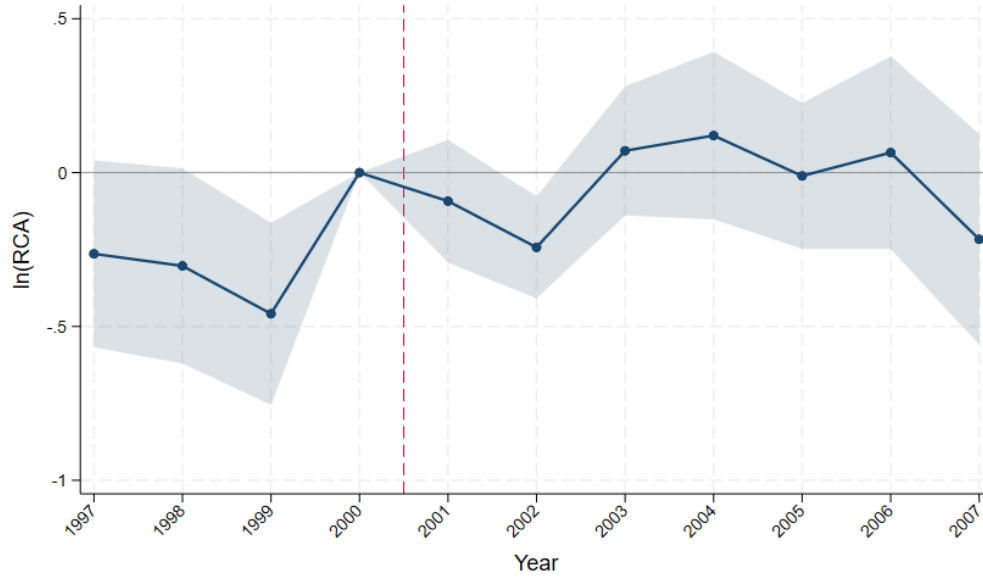


Figure 6: India's Automotive RCA: Event Study

Notes: Figure plots the coefficients β_τ from the regression

$\ln(RCA_{ct}) = \alpha_c + \alpha_t + \sum_{\tau \neq 2000} \beta_\tau \cdot (\text{India}_c \times \mathbf{1}[t = \tau]) + \varepsilon_{ct}$, with 2000 as the omitted base year. The sample includes all countries with available COMTRADE export data, 1997–2007. Error bars represent 95% confidence intervals based on standard errors clustered at the country level. The negative pre-trends are consistent with India's declining automotive competitiveness prior to the TBTs, which motivated the policy intervention. The sharp reversal after 2001 coincides with the firm-level upgrading effects documented in the paper.

Table 1: Details of STCs

STC Item No. (1)	Implementation Year (2)	TBT Year (3)	STC Year (4)	Description (5)	No. of Industries Affected (4-digit) (6)	Comments (7)
54	2001	2001	2001	Labelling Requirements, Registration, Mandatory Certification	64	–
55	2001	Not Implemented	2001	Revisions of 1955 Prevention of Food Adulteration Act	2	Resolved in 2001
70	2001	2001	2002	Time Requirements for Product's Shelf Life	161	–
84	2001	2001	2002	Regulation on Second Hand Vehicles and New Vehicles	10	–
104	2001	2001	2004	Homologation of Vehicles	14	–

Notes: Authors' Compilation from WTO STC Database and Notifications from Gazette of India.

Table 2: Endogeneity of TBTs

	TBT Incidence = 1					
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(\text{Imports})_{2000,j}$	0.004*** (0.001)					
$\Delta \text{Imports}_j$		-0.002 (0.001)				
$\ln(\text{Avg Imports})_j$			0.005*** (0.002)			
$\ln(\text{Markup})_{2000,j}$				0.021 (0.066)		
ΔMarkup_j					0.100 (0.117)	
$\ln(\text{Avg Markup})_j$						-0.070 (0.074)
N	1,241	1,241	1,241	65	64	65
R-squared	0.766	0.765	0.766	0.344	0.347	0.352
Industry (2-digit) FE	✓	✓	✓	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (6) use TBT incidence as the dependent variable. In particular, it takes the value of 1 if an industry j (2004 NIC 4-digit) was treated or affected by a TBT in 2001. Industry level markups are calculated as real sales weighted average of firm-product level markups. Firm-product level markups are calculated following [De Loecker *et al.* \(2016\)](#). Numbers in the parentheses are clustered standard errors at industry (NIC 4-digit) level.

Table 3: TBTs and Import Flows

	Imports					
	OLS			PPML		
	(1)	(2)	(3)	(4)	(5)	(6)
$Treated_{jt}$	-0.463** (0.208)	-0.426** (0.217)	-0.029* (0.017)	-0.012 (0.155)	0.007 (0.163)	0.007 (0.163)
$Treated_{jt} \times Automotive_j$		-0.624 (0.595)	-0.068* (0.036)		-0.356 * (0.185)	-0.356* (0.185)
Automotive TBTs Effect		-1.050* (0.559)	-0.097*** (0.329)		-0.349*** (0.137)	-0.350*** (0.138)
N	9,928	9,928	2,283,440	9,864	9,864	2,101,032
R-squared	0.907	0.907	0.448	–	–	–
Industry FE	✓	✓	✓	✓	✓	✓
Country FE	X	X	✓	X	X	✓
Year FE	✓	✓	✓	✓	✓	✓
Industry FE $\times$ Year trend	✓	✓	✓	✓	✓	✓

Notes: All regressions use 1997–2005. Columns (1)–(3) are OLS with $\ln(\text{imports})$ as the dependent variable; columns (4)–(6) are PPML with imports (levels) as the dependent variable. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_j$ takes a value 1 for the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at HS 4-digit level. Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table 4: TBTs and Technology Adoption Expenses

	Total Technology Adoption Expenses	R&D Expenses			Technology Transfer Expenses		
		Total	Capital	Current	Total	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated _{<i>it</i>}	0.041 (0.044)	0.006 (0.043)	0.006 (0.030)	0.015 (0.042)	0.028 (0.025)	0.030 (0.047)	0.009 (0.045)
Treated _{<i>it</i>} × Automotive _{<i>ij</i>}	0.283*** (0.080)	0.215*** (0.078)	0.190*** (0.062)	0.165** (0.079)	0.259*** (0.064)	0.177*** (0.054)	0.158** (0.063)
Automotive TBTs Effect	0.325*** (0.079)	0.221*** (0.077)	0.196*** (0.061)	0.181** (0.076)	0.287*** (0.067)	0.203*** (0.059)	0.212*** (0.066)
N	22,767	22,767	22,767	22,767	22,767	22,767	22,767
R-squared	0.864	0.854	0.694	0.864	0.790	0.812	0.806
Firm FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (3) use total technology adoption expenses as the dependent variable. Total technology adoption expenses are defined as the sum of total expenses on account of R&D and technology transfer. *Treated_{it}* takes a value 1 for an industry *j* (at NIC 2004 4-digit level) for which India imposed a TBT at time *t*. *Automotive_{ij}* takes a value 1 for firm *i* belonging to *j* which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%

Table 5: Decomposing the Automotive TBT Effect on Technology Adoption: Regulation 84 vs. 104

	Technology Adoption	R&D Expenses			Technology Transfer		
		Total (2)	Capital (3)	Current (4)	Total (5)	Domestic (6)	Foreign (7)
$Treated_{jt}$	0.041 (0.044)	0.006 (0.043)	0.006 (0.030)	0.015 (0.042)	0.028 (0.025)	0.026 (0.048)	0.054 (0.050)
$Treated_{jt} \times Reg. 84_j$	0.209 (0.242)	0.218 (0.241)	0.113 (0.214)	0.255 (0.212)	0.074 (0.223)	0.102 (0.160)	0.131 (0.188)
$Treated_{jt} \times Reg. 104_j$	0.230*** (0.059)	0.159*** (0.032)	0.161*** (0.023)	0.100*** (0.036)	0.240*** (0.042)	0.151*** (0.030)	0.124*** (0.032)
N	22,767	22,767	22,767	22,767	22,767	22,767	22,767
R-squared	0.864	0.855	0.695	0.864	0.790	0.812	0.806
Firm FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓

Notes: All regressions at the firm-year level, 1997–2005. $Treated_{jt}$ takes a value 1 for an industry j for which India imposed a TBT at time t . Reg. 84 covers second-hand vehicle import ban and ARAI testing requirements; Reg. 104 covers ARAI certification for vehicle components. All industries subject to Reg. 84 are also subject to Reg. 104, so the Reg. 84 coefficient captures the *additional* effect of the import ban beyond the certification requirement. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table 6: TBTs and Technology Adoption: Mechanisms – Who is driving the effect?

	Technology Adoption Expenses (1)	R&D Expenses (2)	Technology Transfer Expenses (3)
Panel A: All Characteristics			
$Post_t \times R\&D\ firm_i$	0.047 (0.102)	0.058 (0.034)	0.080 (0.118)
$Post_t \times Large_i$	0.200* (0.103)	0.261*** (0.043)	0.054 (0.082)
$Post_t \times Exporter_i$	0.064 (0.136)	0.110 (0.128)	0.011 (0.086)
$Post_t \times Import\ Dependence_i$	0.484*** (0.066)	0.472*** (0.085)	0.244** (0.084)
$Post_t \times Parts\ maker_i$	-0.062 (0.150)	-0.166 (0.129)	0.013 (0.127)
N	1,841	1,841	1,841
R-squared	0.866	0.838	0.789
Panel B: Different Types of Importers			
	(4)	(5)	(6)
$Post_t \times Importer_{Capital\ goods}$	0.576*** (0.097)	0.572*** 0.122	0.316** 0.093
$Post_t \times Importer_{Raw\ materials}$	0.484*** (0.075)	0.555*** (0.106)	0.198** (0.063)
$Post_t \times Importer_{Stores\ \&\ spares}$	0.412*** (0.117)	0.461** (0.158)	0.184* (0.090)
$Post_t \times Importer_{Finished\ goods}$	0.169* (0.087)	0.192 (0.139)	0.042 (0.083)
N	2,065	2,065	2,065
R-squared	0.867	0.834	0.795
Panel C: Capital Import Dependency			
	Import of Capital Goods (7)	R&D Expenses (Capital) (8)	Domestic Technology Transfer Expenses (9)
$Post_t \times Importer_{Capital\ goods}$	-0.267** (0.113)	0.400*** (0.100)	0.368*** (0.078)
N	2,065	2,065	2,065
R-squared	0.732	0.822	0.611
Firm FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All regressions at the firm-year level, 1997–2005. Sample is restricted to only automotive TBT industries (Reg. 84 & 104). $Post_t$ takes a value 1 if year is > 2001. R&D firm = 1 if average of R&D expenses before 2001 > 0; Large = 1 if average assets of a firm before 2001 > median asset value of the automotive sector; Exporter = 1 if a firm's average exports > 0 before 2001; Import dependence = 1 if a firm's average ratio of total imports to assets is > median ratio of the automotive sector; Parts maker = 1 if a firm belongs to the 2004 NIC 4-digit = 3430. $Importer_{Capital\ goods}$, $Importer_{Raw\ materials}$, $Importer_{Stores\ \&\ spares}$, and $Importer_{Finished\ goods}$ takes a value 1 if a firm's average ratio of import of capital goods, raw materials, stores & spares, and finished goods to total assets before 2001 is above the median value of the ratio of the entire automotive sector. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table 7: TBTs and Input Composition

	Input Scope	Value of Inputs Used		Average Input Expenses
	(1)	(2)	(3)	(4)
Panel A: Aggregate Effects				
$Treated_{jt}$	0.003 (0.017)	0.031 (0.037)	0.018 (0.040)	0.026 (0.052)
$Treated_{jt} \times Automotive_j$	0.055*** (0.017)	0.199*** (0.057)	0.186*** (0.060)	0.221*** (0.067)
Automotive TBTs Effect	0.057** (0.020)	0.230*** (0.057)	0.204*** (0.057)	0.247*** (0.063)
N	18,343	70,980	55,596	18,096
R-squared	0.898	0.923	0.907	0.894
Panel B: Decomposing between Reg. 84 and reg. 104				
	(5)	(6)	(7)	(8)
$Treated_{jt}$	0.003 (0.017)	0.031 (0.037)	0.018 (0.040)	0.026 (0.052)
$Treated_{jt} \times Reg. 84_j$	-0.050 (0.047)	-0.239** (0.097)	-0.273*** (0.095)	-0.235 (0.145)
$Treated_{jt} \times Reg. 104_j$	0.066*** (0.015)	0.263*** (0.055)	0.258*** (0.049)	0.274*** (0.066)
N	18,343	70,980	55,596	18,096
Adj. R-sq	0.874	0.903	0.907	0.869
Controlling for Price	X	X	✓	X
Firm FE	✓	X	✓	✓
Firm-Input FE	X	✓	✓	X
Year FE	✓	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Column (1) use total number of inputs used in the production; columns (2) – (3) use value of each input used; column (4) uses average expenses per input (total expenses on inputs/number of inputs) used by a firm as the dependent variable. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table 8: TBTs and Product Level Outcomes

	Product Scope (1)	Product Entry (2)	Product Exit (3)
Treated _{jt}	0.003 (0.015)	−0.009 (0.013)	−0.004 (0.010)
Treated _{jt} × Automotive _{ij}	0.029* (0.019)	−0.031 (0.020)	−0.027*** (0.007)
Automotive TBTs Effect	0.032 (0.022)	−0.040* (0.021)	−0.031*** (0.011)
N	24,263	91,154	91,154
R-squared	0.875	0.334	0.336
Firm FE	✓	X	X
Firm-Product FE	X	✓	✓
	Price (4)	Markups (5)	Marginal Cost (6)
Treated _{jt}	0.081 (0.050)	0.060 (0.060)	0.021 (0.069)
Treated _{jt} × Automotive _{ij}	0.090 (0.082)	0.235 (0.165)	−0.145 (0.158)
Automotive TBTs Effect	0.172** (0.082)	0.295* (0.161)	−0.123 (0.162)
N	17,579	17,579	17,579
R-squared	0.981	0.161	0.162
Firm-Product FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Column (1) uses total product scope – the total number of products produced; column (2) uses product entry – it takes a value 1 when a new product appears in the list of products produced by a firm given that it has not been produced before; column (3) uses product exit – it takes a value 1 if a product has been dropped from the list of products produced by a firm and not observed hereafter; column (4) uses price of a product; column (5) uses markup of a product; column (6) uses marginal cost of a product produced by a firm as the dependent variable, respectively. Price, Markups, and Marginal cost at the product level are estimated using [De Loecker et al. \(2016\)](#). *Treated_{jt}* takes a value 1 for an industry *j* (at NIC 2004 4-digit level) for which India imposed a TBT at time *t*. *Automotive_{ij}* takes a value 1 for firm *i* belonging to *j* which belong to the automotive industries or affected by the regulations no. 84 and 104. For columns (2) and (3), we report marginal effects. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10% .

Table 9: TBTs, Domestic Sales, and Exports

	Domestic Sales				Exports			Probability of Exporting	
	Domestic Firms		Non-Exporters		Domestic Firms		Non-Exporters		Non-Exporters
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
Treated _{it}	0.109 (0.072)	0.061 (0.086)	0.193 (0.136)	0.014 (0.060)	0.012 (0.063)	0.002 (0.062)	0.007 (0.006)		
Treated _{it} × Automotive _{ij}	0.174** (0.085)	0.156* (0.087)	0.259* (0.143)	0.203** (0.081)	0.164* (0.092)	0.285*** (0.070)	0.049*** (0.017)		
Automotive TBTs Effect	0.283*** (0.087)	0.217** (0.0982)	0.452*** (0.160)	0.217** (0.086)	0.176* (0.095)	0.287*** (0.084)	0.056*** (0.017)		
N	22,767	19,418	7,888	22,767	19,418	7,888	22,767		
R-squared	0.895	0.894	0.839	0.873	0.872	0.437	0.426		
Firm FE	✓	✓	✓	✓	✓	✓	✓		✓
Year FE	✓	✓	✓	✓	✓	✓	✓		✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (3) use domestic sales and columns (4) – (6) use exports of a firm as the dependent variable, respectively. *Treated_{it}* takes a value 1 for an industry *j* (at NIC 2004 4-digit level) for which India imposed a TBT at time *t*. *Automotive_{ij}* takes a value 1 for firm *i* belonging to *j* which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table 10: TBTs and Revealed Comparative Advantage (RCA)

	Revealed Comparative Advantage		
	All	Developing	Advanced
	Countries	Countries	Countries
	(1)	(2)	(3)
$Post_t \times India_c$	0.122*** (0.032)	0.110 (0.086)	0.130*** (0.025)
N	1,189	812	386
Country FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All the regressions are run for 1996–2005. Dependent variable is RCA in levels as we carry out the estimations in PPML. $RCA = (\text{country automotive exports} / \text{country total exports}) / (\text{world automotive exports} / \text{world total exports})$. Automotive exports defined as HS chapter 87 (vehicles, parts & accessories). Sample covers 1997–2005, matching the firm level analysis window. $Post_t = 1$ for years ≥ 2001 . “Developing” restricts the sample to non-high-income economies (World Bank classification). “Advanced” restricts to high-income/OECD economies, with India included for identification. All specifications include country and year fixed effects. Numbers in the parentheses are clustered standard errors at country level. Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Online Appendix for

When a Protectionist Trade Policy turns into an Industrial Policy: Evidence from India

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A Specific Trade Concerns

This section contains details of each of the concern that we exploit and a summary of the legislative developments of the corresponding measures.

A.1 STC 54

Date: 30/03/2001

The concern was raised by Australia, Canada, European Union, Japan, and the United States. Key issues were related to transparency, unnecessary barriers to trade, rationale, legitimacy, further information, and clarification. The regulation was notified in the Gazette of India on 24 November 2000 and entered into force on 2 January 2001 (text available at <http://DGFTCOM.NIC.IN/EXIM/2000/NOT/NOT00/NOT4400.htm>).

The authorities raised concerns about the proposals on the following elements:

- the lack of a reasonable period of time for suppliers to adapt to the new requirements;
- the rationale for requiring labels on non-perishable products to indicate the month and year of packaging as well as the maximum retail price;
- the justification of the additional costs imposed on manufacturers and consumers;
- the reason why certain products were exempted from the labelling requirement;
- the role of the BIS and the reason why manufacturers and exporters must register with BIS;

- the coverage of the labelling requirements (noting that on 22 January 2001, clarification had been made that the labelling requirements applied only to the import of pre-packaged commodities intended for retail sale).

These regulatory proposals had not been notified to the WTO. India notified them ex post through notification G/TBT/N/IND/1 submitted on 08/05/2002 (distributed on 28/05/2002).

Product classification: manually codified.

A.2 STC 55

Date: 30/03/2001

The concern was raised by the United States regarding a proposed TBT to amend the Prevention of Food Adulteration Rules, 1955, by the Health Ministry of the Indian government. The proposed TBT was not notified to the WTO (nor was it later).

Although the amendments targeted various products, the concern related specifically to alcoholic beverages and the permitted levels of residues. When the TBT was finally implemented in 2001, the entire section related to alcoholic beverages was dropped. Hence, the TBT was not trade-restrictive when implemented.

Product classification: STC dataset.

A.3 STC 70

Date: 15/03/2002

The concern was raised by the European Union regarding a TBT measure banning all food products with less than 60% remaining shelf life at the time of import. The measure was enforced on 30 July 2001 (Notification No. 22 (RE-2001) 1997–2002, available at <http://www.nic.in/exinpol>).

It amended Chapter 1A (General Notes regarding Import Policy) of ITC(HS) Classifications of Export and Import Items, 1997–2002. The measure was notified to the WTO only after enforcement, on 10 July 2002, through notification G/TBT/N/IND/8.

Product classification: STC dataset.

A.4 STC 84

Date: 17/10/2002

The concern was raised by the European Union regarding a TBT measure banning imports of second-hand vehicles older than three years. Only certificates issued by Indian authorities were recognized, even though the regulation resembled the UN/ECE standard. The TBT was enforced in 2001 and remained unresolved during the sample period.

The measure was notified on 10 July 2002 (G/TBT/N/IND/9), although it referred to a measure already in force (DGFT Notification no. 4(RE-2001)/97-2992, dated 31 March 2001).

Product classification: STC dataset.

A.5 STC 104

Date: 23/03/2004

The concern was raised by the European Union regarding a TBT requiring mandatory certification from India's approval agency (ARAI) for new vehicles and components. ARAI lacked testing capabilities for large engines, requiring foreign companies to build or borrow testing facilities. The TBT was enforced in 2001 and resolved in 2004 when India accepted certifications from international agencies for components where ARAI lacked testing capacity.

The measure corresponded to DGFT Notification no. 4(RE-2001)/97-2992, dated 31 March 2001. It was notified to the WTO only on 10 July 2002 (G/TBT/N/IND/9).

Product classification: STC dataset.

B Data Description

We use an annual panel of Indian manufacturing firms that covers around 6,000 firms, across 105 industries (4-digit), over the period of 1996–2007. This firm level data is used from the PROWESS database of the Centre for Monitoring Indian Economy (CMIE). All monetary-based variables measured in Millions of Indian Rupees (INR), deflated by 2005 industry-specific Wholesale Price Index (WPI). We use 2004 National Industrial Classification (NIC).

B.1 Variable Definitions

- **Treated:** Dummy variable – it takes a value 1 if a firm belongs to one of the industries targeted by the TBTs
- **Not Treated:** Dummy variable – it takes a value 1 if a firm does not belong to one of the industries targeted by the TBTs
- **Treated once:** Dummy variable – it takes a value 1 if a firm belongs to only one TBTs
- **Treated twice:** Dummy variable – it takes a value 1 if a firm has been affected by two TBTs
- **Treated by either n.54 or n.70 :** Dummy variable – it takes a value 1 if a firm has been affected by regulations 54 or 70
- **Treated by either n.84 or n.104 :** Dummy variable – it takes a value 1 if a firm has been affected by regulations 84 or 104
- **Treated by both n.54 and n.70 :** Dummy variable – it takes a value 1 if a firm has been affected by both regulations 54 and 70
- **Treated by both n.84 and n.104 :** Dummy variable – it takes a value 1 if a firm has been affected by both regulations 84 and 104
- **Post:** Dummy variable – it takes a value of 1 for years after 2001
- **Retail:** Dummy variable – it takes a value 1 if an industry is covered by TBT numbers 54 and 70 (retail industries)
- **Automotive:** Dummy variable – it takes a value 1 if an industry is covered by TBT numbers 84 and 104 (automotive industries)

- **Regulation no. 54:** Dummy variable – it takes a value 1 if an industry is covered by the TBT number 54
- **Regulation no. 70:** Dummy variable – it takes a value 1 if an industry is covered by the TBT number 70
- **Regulation no. 84:** Dummy variable – it takes a value 1 if an industry is covered by the TBT number 84
- **Regulation no. 104:** Dummy variable – it takes a value 1 if an industry is covered by the TBT number 104
- **Sales:** Log of total sales value of a firm
- **Domestic Sales:** Log of domestic sales value of a firm
- **Export:** Log of export value of a firm
- **R&D Capital:** Log of capital R&D expenditures of a firm
- **R&D Current:** Log of current R&D expenditures of a firm
- **Technology Transfer:** Log of expenditure on technology transfer of a firm
- **Total Assets:** Log of value of total assets of a firm
- **Import:** Log of total import value of a firm
- **Product Scope:** Log of the total number of products produced by a firm
- **Input Scope:** Log of the total number of inputs used by a firm in the production process
- **Product Entry:** Dummy variable – it takes a value 1 when a new product appears in the sample of the products produced by a firm in a particular year given that the product never appeared before in that sample
- **Product Exit:** Dummy variable – it takes a value 1 if a product disappears from the sample of the products produced in a given year conditional on that fact that it has been produced at least once in any year before
- **Product Sales:** Log of total sales value at the firm-product level
- **Sales Share:** Share of a product sales in the total sales value of a firm
- **Product Quantity:** Log of total quantity value at the firm-product level

- **Product Price:** Log of unit price (value of a product sales/quantity of product sold) at a firm-product level
- **Input Entry:** Dummy variable – it takes a value 1 if an input is used for the first time by a firm in its production process given that input was not used before
- **Input Exit:** Dummy variable – it takes a value 1 if an input is not found to have been used in a particular year in the sample of inputs given that input has been used at least once before in the production process of a firm
- **Total Input Value:** Log of value of total inputs used by a firm
- **Input Share:** Share of value of a particular input in the total input expenditure of a firm
- **Input Quantity:** Log of quantity purchased of each inputs used in the production process (firm-input level)
- **Input Price:** Log of unit price of an input used (firm-input level)
- **Imports:** Log of Indian import flows in 2000 from the rest of the world (industry level)
- Δ **Imports:** Log of past change in Indian import flows from the rest of the world in 2000 (industry level)
- **Avg Imports:** Log of average import flows from the rest of the world in 1997-2000 (industry level)
- **Markup:** Log of markup in 2000 (industry level)
- Δ **Markup:** Log of past change in markup in 2000 (industry level)
- **Avg Markup:** Log of average markup in 1997-2000 (industry level)

C Tables

C.1 Descriptive Statistics

Table C.1: Descriptive Statistics: Input Use and Products Produced by Different Groups of Firms

	Mean	Std. Dev.	Min	Max
Panel A: Number of Inputs Used				
Aggregate	4.221	0.344	3.748	4.807
Firms with Automotive TBTs	4.138	0.171	3.890	4.350
Firms with Retail TBTs	3.882	0.081	3.748	3.986
Firms with No TBT	4.643	0.142	4.422	4.807
Panel B: Value of Inputs Used				
Aggregate	346.880	277.028	131.94	1304.65
Firms with Automotive TBT	517.76	252.93	262.46	1026.70
Firms with Retail TBTs	315.07	353.11	131.94	1304.65
Firms with No TBT	207.81	59.80	141.54	333.94
Panel C: Number of Products Produced				
Aggregate	4.728	0.206	4.470	5.284
Firms with Automotive TBTs	4.638	0.119	4.470	4.835
Firms with Retail TBTs	4.639	0.115	4.487	4.776
Firms with No TBT	4.906	0.237	4.637	5.284
Panel D: Value of Products Produced				
Aggregate	495.72	271.75	235.58	1301.16
Firms with Automotive TBTs	797.57	266.30	584.56	1301.16
Firms with Retail TBTs	315.42	64.93	235.58	451.26
Firms with No TBT	374.16	95.10	256.02	565.80

Notes: The table reports summary statistics of the number and value of inputs used by firms, by treatment category. Panel A refers to the average number of inputs, while Panel B refers to the average unit value of inputs (in INR). Treatment groups are defined based on exposure to different types of TBT events.

Table C.2: Summary Statistics

	Mean (1)	Std. Dev. (2)	Obs. (3)
Panel A: Firm Level Variables:			
Sales	5.068	2.430	23,393
Domestic Sales	4.854	2.454	23,393
Export Value	1.776	2.372	23,393
R&D Capital	0.162	0.651	23,393
R&D Current	0.374	0.967	23,393
Foreign Technology Transfer	0.238	0.803	23,393
Total Assets	5.538	1.806	23,393
Import Value	2.037	2.334	23,393
Product Scope	1.168	0.791	22,127
Input Scope	1.117	0.780	18,787
Total Value of Inputs	4.939	1.920	18,539
Panel B: Firm-Product Level Variables:			
Product Entry	0.140	0.347	134,311
Product Exit	0.107	0.310	134,311
Product Sales	3.308	2.275	115,182
Product Level Sales Share	0.249	0.346	115,185
Product Quantity Sold	4.790	3.746	73,993
Product Price	-1.308	3.365	69,700
Panel C: Firm-Input Level Variables:			
Input Entry	0.051	0.221	126,059
Input Exit	0.054	0.226	126,059
Total Value of Inputs	3.079	1.957	74,989
Share of an Input Purchased	0.248	0.304	74,891
Input Quantity Purchase	8.450	3.700	63,781
Input Price	-5.825	3.489	59,135

Notes: Table reports values for 1996–2007 in logarithm terms except for product and input scope. Those are reported as numbers.

Table C.3: Firms in Treated and Control Groups

Category	# Firms	# NIC 2004
Treated	3,368	51
Not treated	1,627	48
Treated once	2,809	37
Treated twice	559	14
Treated by either n.54 or n.70	2,954	42
Treated by either n.84 or n.104	414	9
Treated by both n.54 and n.70	454	7
Treated by both n.84 and n.104	105	7
Total	4,995	99

Notes: Authors' computation from the matching procedure.

Table C.4: Balance Tests: Firms in Treated and Control Groups

	No TBT	TBT	Difference
	(1)	(2)	(3)
R&D (capital)	0.134 (0.551)	0.179 (0.661)	0.046 (0.046)
R&D (current)	0.346 (0.892)	0.375 (0.929)	0.029 (0.081)
Technology Transfer	0.304 (0.889)	0.258 (0.804)	−0.046 (0.068)
Total Sales	5.242 (2.229)	5.164 (2.252)	−0.077 (0.174)
Domestic Sales	4.947 (2.357)	4.990 (2.282)	0.043 (0.233)
Exports	1.915 (2.330)	1.701 (2.263)	−0.214 (0.204)
Size (Assets)	5.783 (1.658)	5.636 (1.631)	−0.147 (0.131)
Age	20.982 (17.138)	20.293 (17.072)	−0.689 (1.483)

Notes: Pre-treatment means (1997–2001). Column (3) reports mean differences; standard errors in parentheses.

Table C.5: TBTs and Firm Entry-Exit

	Active (1)	Dissolved (2)	Dormant (3)	Amalgamated (4)
Treated_{j-2001}	-0.018 (0.021)	0.014 (0.012)	0.006 (0.007)	-0.002 (0.011)
$\text{Treated}_{j-2001} \times \text{Automotive}_j$	0.068*** (0.019)	-0.024 (0.021)	0.005 (0.006)	-0.050*** (0.008)
Automotive TBTs Effect	0.050** (0.024)	-0.011 (0.022)	0.002 (0.007)	-0.051*** (0.011)
N	3,356	3,356	3,356	3,356
R-squared	0.002	0.001	0.000	0.003
Industry (4-digit) FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: All regressions are cross-sectional at the firm level in 2001. The dependent variable is a dummy variable that takes the value of 1 if the company is classified as Active, Dissolved, Dormant or Amalgamated in the business registry in 2015. NIC 4-digit fixed effects absorbed. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

C.2 Endogeneity

Table C.6: Endogeneity of TBTs: Markup Robustness (Mean, 75th, and 90th Percentile)

	TBT Incidence = 1		
	Mean Markup		
	(1)	(2)	(3)
$\ln(\text{markup})_{t-1}$	0.021 (0.066)		
$\Delta \ln(\text{markup})_{t-1}$		0.100 (0.117)	
$\text{Avg}(\text{markup})_{Pre-2001}$			-0.070 (0.074)
R-squared	0.344	0.347	0.352
	75th Pctile Markup		
	(4)	(5)	(6)
$\ln(\text{markup})_{t-1}$	-0.058 (0.060)		
$\Delta \ln(\text{markup})_{t-1}$		-0.060 (0.098)	
$\text{Avg}(\text{markup})_{Pre-2001}$			-0.086 (0.077)
R-squared	0.352	0.344	0.356
	90th Pctile Markup		
	(7)	(8)	(9)
$\ln(\text{markup})_{t-1}$	-0.078 (0.060)		
$\Delta \ln(\text{markup})_{t-1}$		-0.078 (0.081)	
$\text{Avg}(\text{markup})_{Pre-2001}$			-0.083 (0.070)
R-squared	0.361	0.348	0.358
N	65	64	65
Industry (2-digit) FE	✓	✓	✓

Notes: All the regressions are run for 1997–2001. Dependent variable is TBT dummy (=1 if a 4-digit industry received a TBT). All regressions are cross-sectional at the NIC 4-digit industry level in 2001. Within each column group, the three columns test lagged level, lagged growth (Δ), and pre-period (1997–2000) average of log markup. Columns (1) – (3) use the industry mean markup (replicating cols (5) – (6) of **Table 2**); columns (4) – (6) use the 75th-percentile markup; columns (7) – (9) use the 90th-percentile markup. Markups are computed at the firm-product level following [Singh & Chanda \(2021\)](#) and collapsed to the NIC 4-digit industry-year level. NIC 2-digit fixed effects absorbed. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

C.3 Import Flows

Table C.7: TBTs and Import Flows – Heterogeneous Effects

	ln(Imports)		
	(1)	(2)	(3)
Treated _{<i>jt</i>}	0.110*** (0.036)	0.070*** (0.026)	0.037* (0.021)
Treated _{<i>jt</i>} × High Income _{<i>c</i>}	−0.269*** (0.065)		
Treated _{<i>jt</i>} × OECD _{<i>c</i>}		−0.720*** (0.165)	
Treated _{<i>jt</i>} × Complainant _{<i>c</i>}			−0.887*** (0.201)
N	2,283,440	2,283,440	2,283,440
R-squared	0.448	0.449	0.448
Industry FE	✓	✓	✓
Year FE	✓	✓	✓
Country FE	✓	✓	✓
Industry FE × Year trend	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (3) use total imports as the dependent variable. $TBT_{ij,t}$ takes a value 1 if India imposed a TBT on industry j (defined at HS 4 digit level) at time t . High Income_{*c*}, OECD_{*c*}, and Complainant_{*c*} are country level dummies. High Income_{*c*} and OECD_{*c*} takes a value 1 if a country c is mentioned as a high-income country and belong to the OECD list according to the WTO country groupings, respectively. Complainant_{*c*} takes a value 1 if a country has raised a complaint or expressed concerns in the WTO against the TBTs imposed by India. Numbers in the parentheses are clustered standard errors at HS 4-digit level. Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

C.4 Technology Adoption

Table C.8: TBTs and Technology Adoption Expenses: Heterogeneity

	Technology Adoption Expenses	
	(1)	(2)
Panel A: Ownership		
	Domestic	Foreign
Treated _{jt}	0.077* (0.030)	-0.218 (0.144)
Treated _{jt} × Automotive _{ij}	0.188*** (0.064)	0.750* (0.416)
Automotive TBTs Effect	0.265*** (0.062)	0.533 (0.423)
N	19,418	1,805
R-squared	0.861	0.855
Panel B: Exporting Status		
	Exporters	Non-exporters
Treated _{jt}	0.063 (0.060)	0.027 (0.030)
Treated _{jt} × Automotive _{ij}	0.341*** (0.107)	0.166* (0.088)
Automotive TBTs Effect	0.404*** (0.101)	0.193** (0.091)
N	11,353	7,888
R-squared	0.882	0.838
Panel C: Technology Capacity		
	> Mean	< Mean
Treated _{jt}	0.110 (0.093)	-0.002 (0.022)
Treated _{jt} × Automotive _{ij}	0.385*** (0.113)	0.065* (0.033)
Automotive TBTs Effect	0.495*** (0.108)	0.063* (0.035)
N	11,116	11,651
R-squared	0.934	0.843
Firm FE	✓	✓
Year FE	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (2) use total technology adoption expenses of a firm as the dependent variable. *Treated_{jt}* takes a value 1 for an industry *j* (at NIC 2004 4-digit level) for which India imposed a TBT at time *t*. *Automotive_{ij}* takes a value 1 for firm *i* belonging to *j* which belong to the automotive industries or affected by the regulations no. 84 and 104. Exporting status is defined if a firm's average exports before 2001 (1997–2000) is greater than zero or not. Technological capacity of a firm is defined as the sum of R&D and technology transfer expenses before 2001. If a firm's average technology adoption expenses is greater than the mean technological expenses of all the firms put together, that firm is identified as having technological capacity above mean or otherwise. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.9: TBTs and Technology Adoption: Alternative Control Group – Automotive vs. Untreated Firms Only

	Technology Adoption Expenses (1)	R&D Expenses (2)	Technology Transfer Expenses (3)
$Treated_{jt}$	0.324*** (0.079)	0.219*** (0.077)	0.287*** (0.067)
N	9,363	9,363	9,363
R-squared	0.880	0.863	0.824
Firm FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (3) use total technology adoption expenses, R&D expenses, and technology transfer of a firm respectively, as the dependent variable. The estimations drop all firms in retail-TBT industries (regulations 54 and 70), retaining only automotive firms and untreated firms as the control group (55 clusters). After dropping retail-TBT firms, automotive firms are the only treated group, so $Treated_{jt}$ directly captures the automotive TBT effect and no interaction term is needed. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.10: TBTs and Technology Adoption – Controlling for Output and Input Tariffs

	Technology Adoption (1)	R&D Expenditures (2)	Technology Transfers (3)
$Treated_{jt}$	0.022 (0.045)	-0.017 (0.043)	0.030 (0.027)
$Treated_{jt} \times Automotive_{ij}$	0.298*** (0.079)	0.229*** (0.078)	0.261*** (0.064)
Automotive TBTs Effect	0.320*** (0.079)	0.212*** (0.077)	0.292*** (0.067)
N	20,412	20,412	20,412
R-squared	0.868	0.859	0.790
Input Tariffs (4-digit)	✓	✓	✓
Output Tariffs (4-digit)	✓	✓	✓
Industry FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (3) use total technology adoption expenses, R&D total expenditures, and technology transfer respectively as the dependent variable. All specifications additionally control for 4-digit industry-level input and output tariffs. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry (NIC 4-digit) level. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.11: TBTs and Technology Adoption: Within Automotive Group Test – Using Reg. 84

	Technology Adoption Expenses (1)	R&D Expenses (2)	Technology Transfer Expenses (3)
$Post_t \times Reg. 84_{ij}$	0.217 (0.261)	0.226 (0.258)	0.080 (0.243)
N	2,065	2,065	2,065
R-squared	0.861	0.825	0.791
Firm FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (3) use total technology adoption expenses, R&D expenses, and technology transfer of a firm respectively, as the dependent variable. $Post_t$ takes a value 1 if year is > 2001 . $Reg.84_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries which are affected by only the regulation no. 84. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.12: TBTs and Technology Adoption: Within-Automotive Heterogeneity – Pre-Treatment Characteristics

	Technology Adoption Expenses (1)	R&D Expenses (2)	Technology Transfer Expenses (3)
Panel A: R&D Intensity			
$Post_t \times R\&D\ firm_i$	0.390*** (0.079)	0.448*** (0.127)	0.217** (0.070)
N	2,065	2,065	2,065
R-squared	0.863	0.830	0.793
Panel B: Firm Size			
$Post_t \times Large_i$	0.443*** (0.116)	0.527*** (0.131)	0.197** (0.079)
N	2,065	2,065	2,065
R-squared	0.864	0.832	0.792
Panel C: Export Status			
$Post_t \times Exporter_i$	0.282* (0.135)	0.360* (0.156)	0.126 (0.087)
N	1,841	1,841	1,841
R-squared	0.859	0.828	0.787
Panel D: Import Dependence			
$Post_t \times Import\ Dependence_i$	0.584*** (0.087)	0.596*** (0.127)	0.299*** (0.072)
N	2,065	2,065	2,065
R-squared	0.867	0.835	0.794
Panel E: Parts Makers vs. Vehicle Assemblers			
$Post_t \times Parts\ maker_i$	-0.015 (0.172)	-0.125 (0.155)	0.041 (0.141)
N	2,065	2,065	2,065
R-squared	0.860	0.825	0.791
Firm FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All regressions at the firm-year level, 1997–2005. Sample is restricted to only automotive TBT industries (Reg. 84 & 104). $Post_t$ takes a value 1 if year is > 2001. R&D firm = 1 if average R&D expenses before 2001 > 0; Large = 1 if average assets of a firm before 2001 > median asset value of the automotive sector; Exporter = 1 if a firm's average exports > 0 before 2001; Import dependence = 1 if a firm's average ratio of total imports to assets is > median ratio of the automotive sector; Parts maker = 1 if 2004 NIC 4-digit = 3430. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.13: Mechanisms: Capital-Dependence (Additional Outcomes)

	R&D Expenses. (Current) (1)	Foreign Technology Transfer Expenses (2)
$\text{Post}_t \times \text{Importer}_{\text{Capital goods}}$	0.464** (0.146)	0.410*** (0.090)
N	2,065	2,065
R-squared	0.852	0.832
Firm FE	✓	✓
Year FE	✓	✓

Notes: All regressions at the firm-year level, 1997–2005. Sample restricted to automotive firms. $\text{Importer}_{\text{Capital goods}}$ takes a value 1 if a firm's average ratio of import of capital goods to total assets before 2001 is above the median value of the ratio of the entire automotive sector. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.14: Mechanisms: Import-Dependence Other Categories

	Import of Capital Goods	R&D Expenses. (Capital)	Domestic Technology Transfer Expenses
	(1)	(2)	(3)
Panel A: Raw Materials Dependence			
$\text{Post}_t \times \text{Importer}_{\text{Raw materials}}$	0.021 (0.139)	0.473*** (0.083)	0.270** (0.083)
N	2,065	2,065	2,065
R-squared	0.730	0.615	0.821
Panel B: Stores & Spares Dependence			
$\text{Post}_t \times \text{Importer}_{\text{Stores\&Spares}}$	-0.088 (0.098)	0.245 (0.158)	0.175* (0.089)
N	2,065	2,065	2,065
R-squared	0.730	0.605	0.819
Panel C: Finished Goods Dependence			
$\text{Post}_t \times \text{Importer}_{\text{Finished goods}}$	-0.398** (0.150)	-0.065 (0.193)	0.174 (0.137)
N	2,065	2,065	2,065
R-squared	0.731	0.602	0.819
Firm FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All regressions are at the firm-year level, 1997–2004. Sample restricted to automotive firms. Each panel reports a separate regression specification. $\text{Importer}_{\text{Raw materials}}$, $\text{Importer}_{\text{Stores\&Spares}}$, and $\text{Importer}_{\text{Finished goods}}$ take a value 1 if a firm's average pre-2001 ratio of the corresponding import category to total assets is above the median value of that ratio in the automotive sector. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

C.5 Input Composition

Table C.15: TBTs and Input Composition: Heterogeneity

	Input Scope (1)	Value of Inputs Used (2)	Input Scope (3)	Value of Inputs Used (4)
Panel A: Ownership				
	Domestic		Foreign	
Treated _{jt}	0.000 (0.017)	0.013 (0.036)	−0.036 (0.018)	−0.033 (0.079)
Treated _{jt} × Automotive _{ij}	0.063*** (0.019)	0.211*** (0.058)	0.031 (0.090)	0.060 (0.095)
Automotive TBTs Effect	0.064*** (0.022)	0.224*** (0.060)	−0.005 (0.097)	0.027 (0.090)
N	16,011	60,137	1,481	7,251
R-squared	0.898	0.922	0.882	0.935
Panel B: Exporting Status				
	Exporters		Non-exporters	
Treated _{jt}	0.003 (0.023)	0.015 (0.035)	0.014 (0.032)	0.146** (0.068)
Treated _{jt} × Automotive _{ij}	0.031 (0.024)	0.171*** (0.052)	0.084*** (0.031)	0.210** (0.083)
Automotive TBTs Effect	0.034 (0.030)	0.186*** (0.052)	0.098** (0.038)	0.356*** (0.095)
N	9,343	42,665	4,978	17,724
R-squared	0.886	0.923	0.877	0.910
Panel C: Technology Capacity				
	> Mean		< Mean	
Treated _{jt}	0.038* (0.021)	0.030 (0.047)	−0.022 (0.025)	0.073 (0.047)
Treated _{jt} × Automotive _{ij}	0.034 (0.024)	0.176*** (0.058)	0.063** (0.028)	0.172* (0.091)
Automotive TBTs Effect	0.072*** (0.027)	0.206*** (0.059)	0.042 (0.034)	0.246*** (0.091)
N	8,407	38,207	7,805	28,441
R-squared	0.896	0.935	0.881	0.900
Firm FE	✓	X	✓	X
Firm-Input FE	X	✓	X	✓
Year FE	✓	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) and (3) use total number of inputs used; columns (2) and (4) use value of an individual input used by a firm as the dependent variable, respectively. *Treated_{jt}* takes a value 1 for an industry *j* (at NIC 2004 4-digit level) for which India imposed a TBT at time *t*. *Automotive_{ij}* takes a value 1 for firm *i* belonging to *j* which belong to the automotive industries or affected by the regulations no. 84 and 104. Exporting status is defined if a firm's average exports before 2001 (1997–2000) is greater than zero or not. Technological capacity of a firm is defined as the sum of R&D and technology transfer expenses before 2001. If a firm's average technology adoption expenses is greater than the mean technological expenses of all the firms put together, that firm is identified as having technological capacity above mean or otherwise. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.16: TBTs and Input Composition: Controlling for Output and Input Tariffs

	Input Scope	Value of	Average Input
	Scope	Inputs Used	Expenses
	(1)	(2)	(3)
$Treated_{jt}$	0.003 (0.017)	0.031 (0.037)	0.027 (0.052)
$Treated_{jt} \times Automotive_{ij}$	0.054*** (0.017)	0.199*** (0.057)	0.221*** (0.067)
Automotive TBTs Effect	0.057*** (0.020)	0.230*** (0.057)	0.248*** (0.063)
N	18,509	71,002	18,262
R-squared	0.899	0.923	0.897
Input tariffs (4-digit)	✓	✓	✓
Output tariffs (4-digit)	✓	✓	✓
Firm-Input FE	X	X	✓
Firm FE	✓	✓	X
Year FE	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Column (1) uses total number of inputs used in the production; column (2) uses value of an individual input used in the production; column (3) uses average expenses per input (total expenses on inputs/number of inputs) used in the production by a firm as the dependent variable, respectively. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

C.6 Product Portfolio

Table C.17: TBTs and Product Portfolio: Controlling for Input and Output tariffs

	Product Scope (1)	Product Entry (2)	Product Exit (3)
$Treated_{jt}$	0.003 (0.015)	−0.009 (0.013)	−0.004 (0.010)
$Treated_{jt} \times Automotive_{ij}$	0.029* (0.017)	−0.031 (0.020)	−0.027*** (0.007)
Automotive TBTs Effect	0.032 (0.019)	−0.040* (0.021)	−0.031*** (0.011)
N	24,330	91,155	91,155
R-squared	0.875	0.334	0.336
Input tariffs (4-digit)	✓	✓	✓
Output tariffs (4-digit)	✓	✓	✓
Firm-Product FE	X	✓	✓
Firm FE	✓	X	X
Year FE	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Column (1) uses total product scope – the total number of products produced; column (2) uses product entry – it takes a value 1 when a new product appears in the list of products produced by a firm given that it has not been produced before; column (3) uses product exit – it takes a value 1 if a product has been dropped from the list of products produced by a firm and not observed hereafter as the dependent variable, respectively. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries or affected by the regulations no. 84 and 104. For column (1), we report marginal effects. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

C.7 Sales and Exports

Table C.18: Decomposing the Automotive TBT Effect on Sales: Regulation 84 vs. 104

	Domestic Sales (1)	Exports (2)
$Treated_{jt}$	0.109 (0.072)	0.014 (0.060)
$Treated_{jt} \times \text{Reg. 84}_j$	-0.180 (0.193)	-0.136 (0.204)
$Treated_{jt} \times \text{Reg. 104}_j$	0.220*** (0.063)	0.238*** (0.067)
N	22,767	22,767
R-squared	0.870	0.843
Firm FE	✓	✓
Year FE	✓	✓

Notes: All the regressions are run for 1997–2005. $Treated_{jt}$ takes a value 1 for an industry j for which India imposed a TBT at time t . Reg. 84 covers second-hand vehicle import ban and ARAI testing requirements; Reg. 104 covers ARAI certification for vehicle components. All industries subject to Reg. 84 are also subject to Reg. 104, so the Reg. 84 coefficient captures the *additional* effect of the import ban beyond the certification requirement. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.19: Alternative Control Group: Automotive vs. Untreated Firms Only

	Domestic Sales	Exports
	(1)	(2)
$Treated_{jt}$	0.283*** (0.088)	0.218** (0.086)
N	9,363	9,363
R-squared	0.901	0.881
Firm FE	✓	✓
Year FE	✓	✓

Notes: All the regressions are run for 1997–2005. Both the regressions drop all firms in retail-TBT industries (regulations 54 and 70), retaining only automotive firms and untreated firms as the control group (55 clusters). After dropping retail-TBT firms, automotive firms are the only treated group, so $Treated_{jt}$ directly captures the automotive TBT effect and no interaction term is needed. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry level (NIC 2004 4-digit level). Intercepts are not reported. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

Table C.20: Effect on Sales: Controlling for Output and Input Tariffs

	Total Sales (1)	Domestic Sales (2)	Exports (3)
$Treated_{jt}$	0.147* (0.078)	0.138* (0.078)	0.011 (0.063)
$Treated_{jt} \times Automotive_{ij}$	0.150 (0.091)	0.153* (0.088)	0.193** (0.081)
Automotive TBTs Effect	0.297*** (0.092)	0.291*** (0.089)	0.204** (0.088)
N	20,412	20,412	20,412
R-squared	0.894	0.891	0.872
Input tariffs (4-digit)	✓	✓	✓
Output tariffs (4-digit)	✓	✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: All the regressions are run for 1997–2005. Columns (1) – (7) use respectively total technology adoption expenses, R&D capital, R&D current, technology transfer, total sales, domestic sales, and exports as the dependent variable. All specifications additionally control for 4-digit industry-level input and output tariffs. $Treated_{jt}$ takes a value 1 for an industry j (at NIC 2004 4-digit level) for which India imposed a TBT at time t . $Automotive_{ij}$ takes a value 1 for firm i belonging to j which belong to the automotive industries or affected by the regulations no. 84 and 104. Numbers in the parentheses are clustered standard errors at industry (NIC 4-digit) level. ***, **, * denotes statistical significance at 1%, 5%, and 10%.

C.8 Other tables

Table C.21: Welfare Decomposition (Using Pre-treatment Averages 1997–2000): Back-of-the-Envelope Estimation

	INR million	USD million
<i>Panel A: Pre-treatment annual averages</i>		
Domestic sales (automotive firms)	464,202	11,052
Exports (automotive firms)	25,262	601
Technology adoption spending	5,488	131
Automotive imports (COMTRADE)	16,316	389
<i>Panel B: Annual welfare components</i>		
Consumer cost (19% × domestic sales)	88,198	2,100
Producer profit gain (20% margin)	31,850	758
Domestic market (33% × sales × margin)	30,637	729
Exports (24% × exports × margin)	1,213	29
Technology investment (38% × tech spending)	2,085	50
<i>Panel C: Summary ratios</i>		
PS/CS ratio (B/A)		0.36
Sensitivity: 10% margin		0.18
Sensitivity: 30% margin		0.54
Producer revenue gain / Consumer cost		1.81

Notes: All percentage effects are computed as $\exp(\hat{\beta}) - 1$ from the Automotive TBTs Effect (sum of $\hat{\beta}_1 + \hat{\beta}_2$) in each source table. Coefficients from Tables 3 (imports: −30%, PPML col. (5)), 4 (technology adoption: +38%, col. (1)), 9 (domestic sales: +33%, col. (1); exports: +24%, col. (4)), and 8 (prices: +19%, col. (6)). Profit margin assumed at 20% (baseline). USD conversion at INR 42/USD (1997–2000 average). Firm-level data from PROWESS (million INR); import data from COMTRADE via WITS (thousands USD, converted).

D Figures

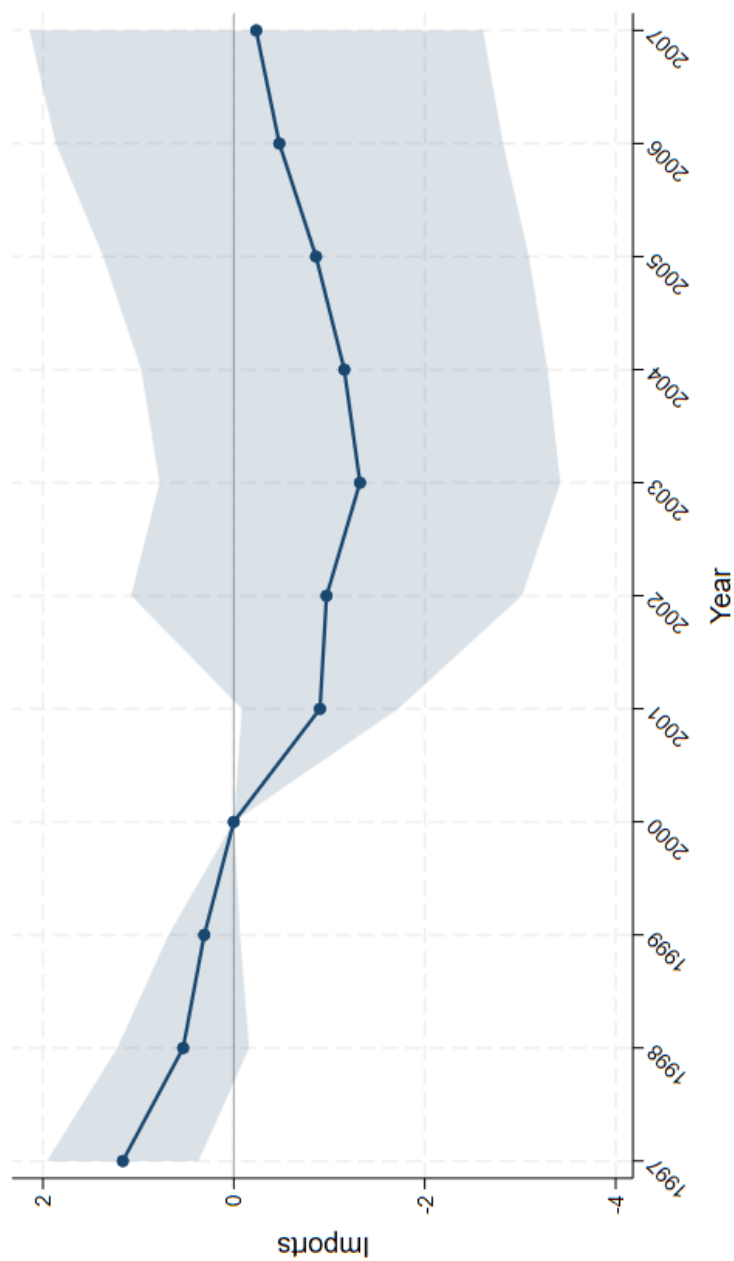


Figure D.1: TBTs and Its Impact on Aggregate Import Flows

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the import flows from RoW of industries (defined at HS 4-digit level) affected by automotive TBT with respect to all other industries. All other firms include industries (i) which are treated with the other regulation, i.e., retail TBT, and (ii) firms which are not affected by any TBT.

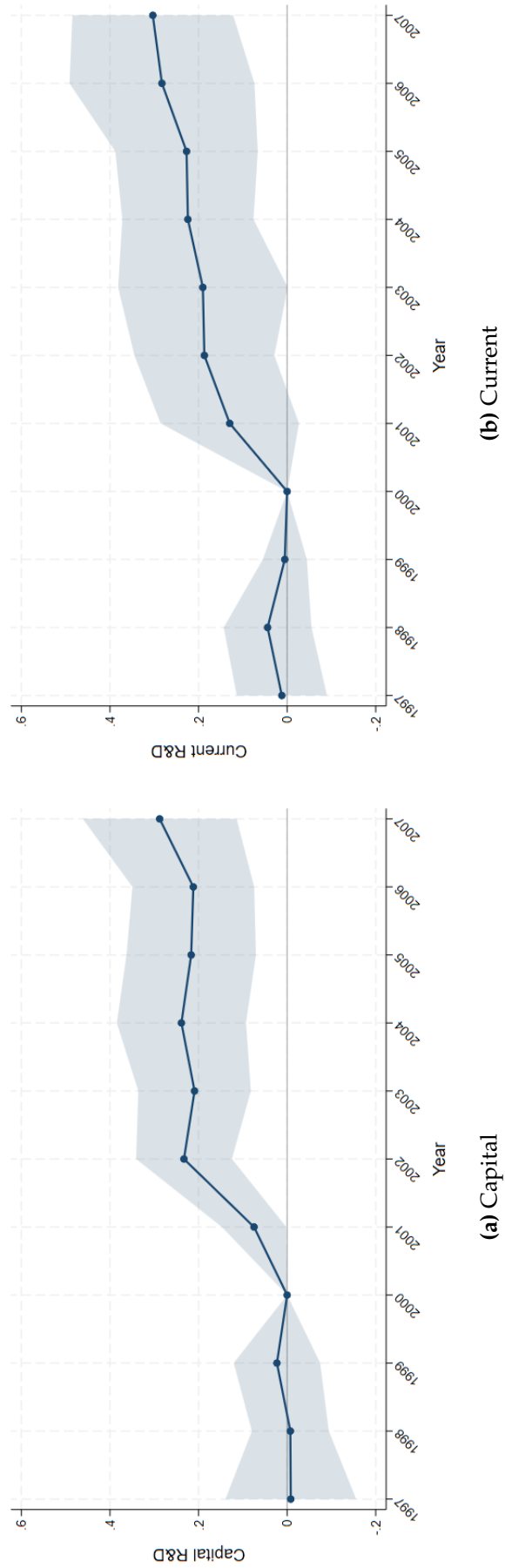
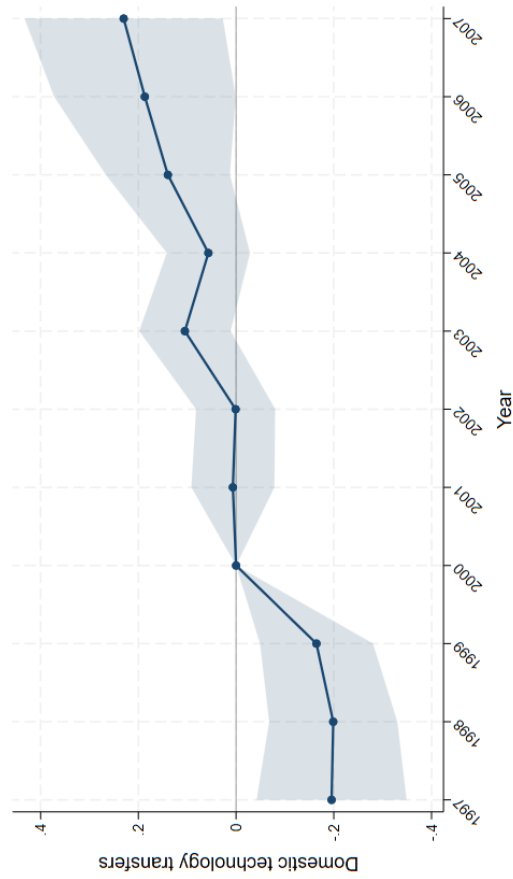
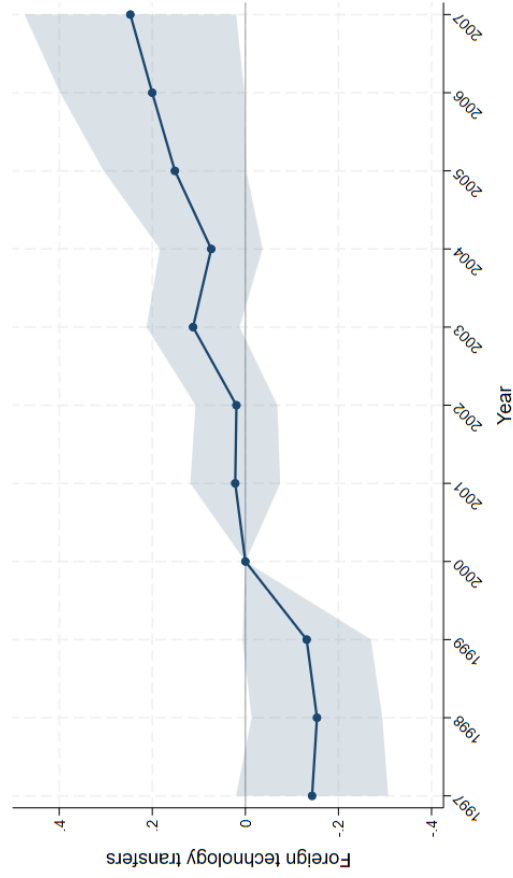


Figure D.2: TBTs and R&D expenses: Capital and Current

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the capital R&D expenses (Panel A) and current R&D expenses (Panel B) between the firms which are affected by the automotive TBTs (regulations no. 84 and 104) and all other firms. All other firms include (i) firms which are treated with the other regulations, i.e., retail TBTs, and (ii) firms which are not affected by any TBT.



(a) Domestic



(b) Foreign

Figure D.3: Automotive TBTs and Effect on Technology Transfers: Domestic and Foreign

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the domestic technology transfer expenses (Panel A), and foreign technology transfer expenses (Panel B) between the firms which are affected by the automotive TBTs (regulations no. 84 and 104) and all other firms. All other firms include (i) firms which are treated with the other regulations, i.e., retail TBTs, and (ii) firms which are not affected by any TBT.

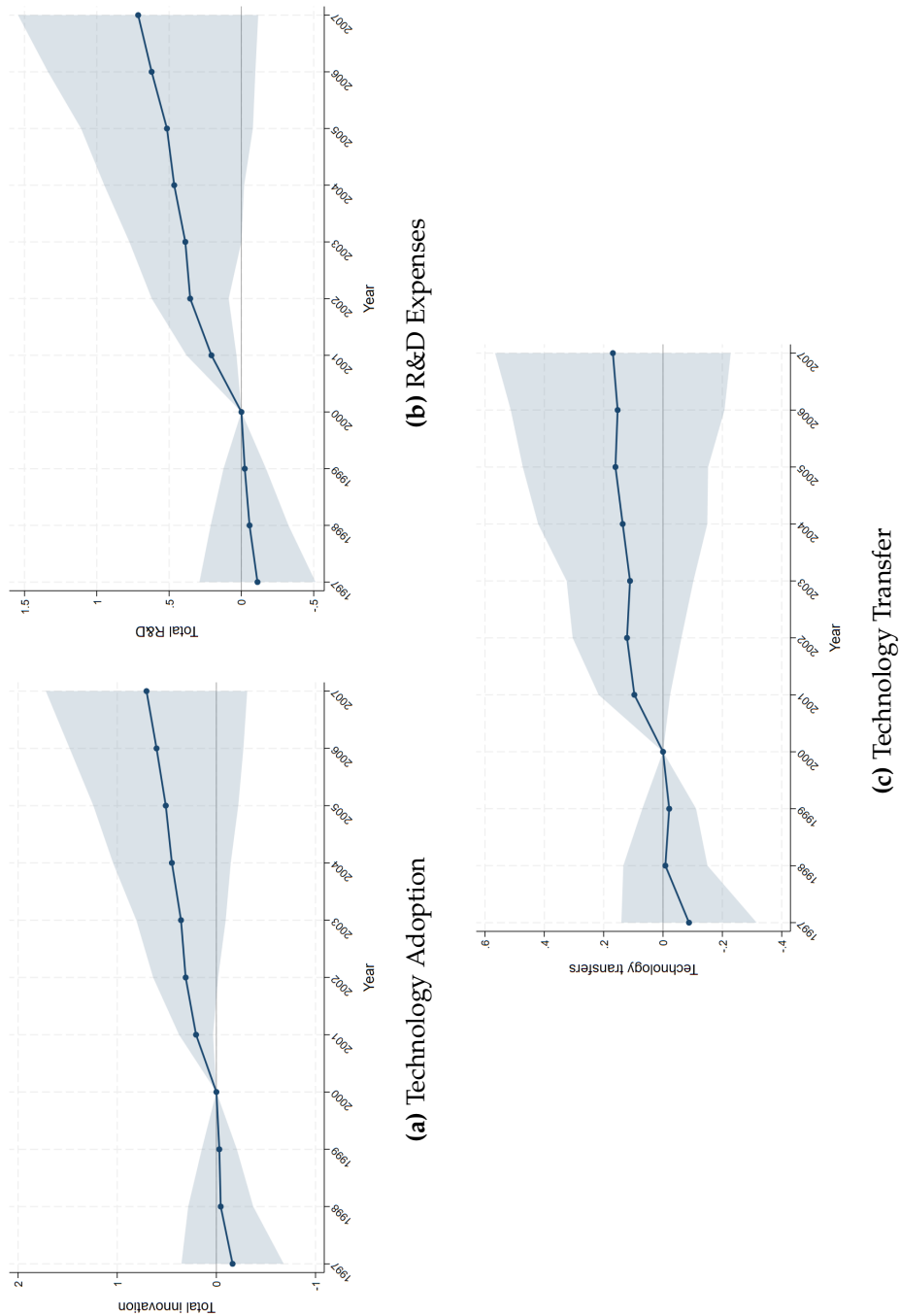
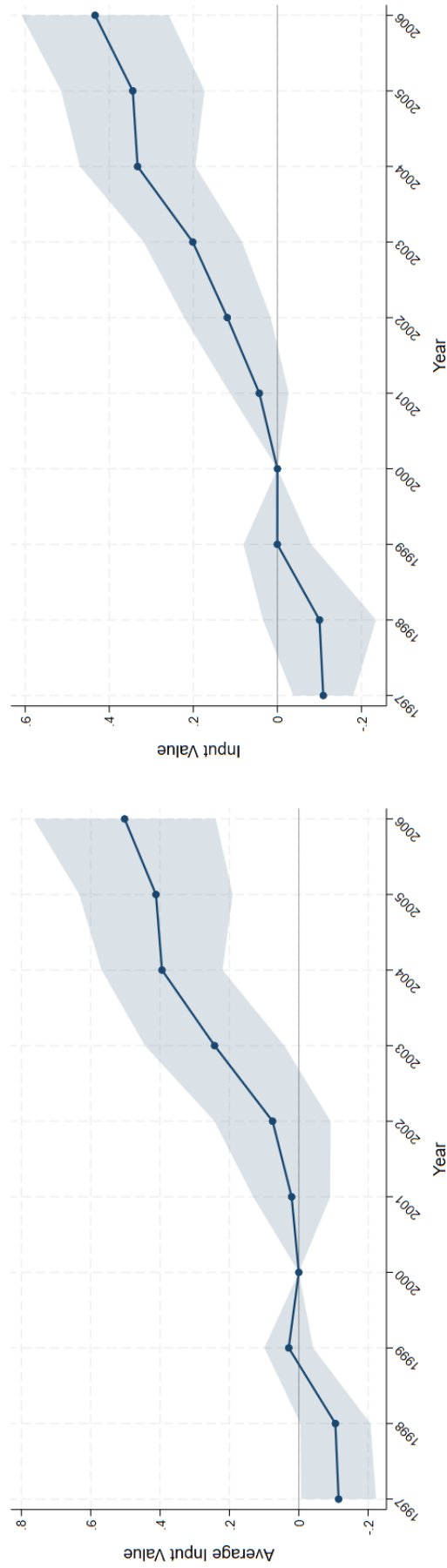


Figure D.4: Automotive TBTs and Technology Adoption, R&D, and Technology Transfer Expenses – Controlling for Time Trends

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) in the Total Technology Adoption expenses (Panel A), R&D expenses (Panel B), and Technology Transfer expenses (Panel C) of firms affected by automotive TBT with respect to all other firms. All other firms include firms (i) which are treated with the other regulation, i.e., retail TBT, and (ii) firms which are not affected by any TBT. All specifications absorb firm fixed effects, year fixed effects, and industry-specific linear trends (NIC 4-digit $\times$ year)

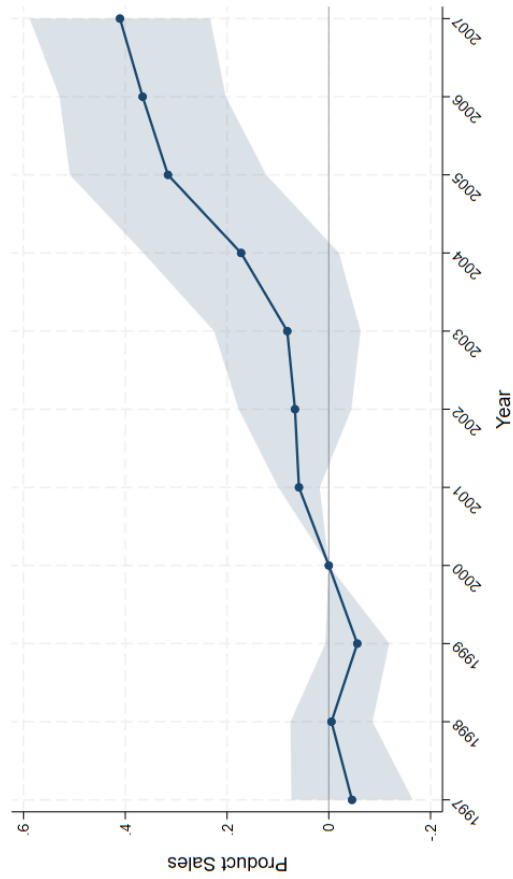


(a) Average Input Value

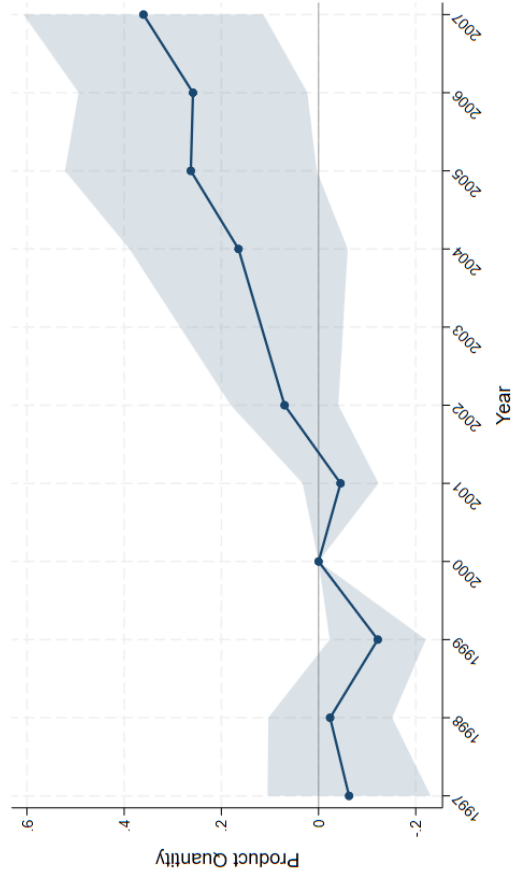
(b) Value per Input

Figure D.5: Automotive TBTs and Effect on Input Value: Domestic Firms

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the firm average input value (Panel A), and value per input (Panel B) between the firms which are affected by the automotive TBTs (regulations no. 84 and 104) and all other firms. All other firms include (i) firms which are treated with the other regulations, i.e., retail TBTs, and (ii) firms which are not affected by any TBT.



(a) Product Sales



(b) Product Quantity

Figure D.6: Automotive TBTs and Effect on Product Portfolio

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the firm product sales value (Panel A), and product quantity (Panel B) between the firms which are affected by the automotive TBTs (regulations no. 84 and 104) and all other firms. All other firms include (i) firms which are treated with the other regulations, i.e., retail TBTs, and (ii) firms which are not affected by any TBT.

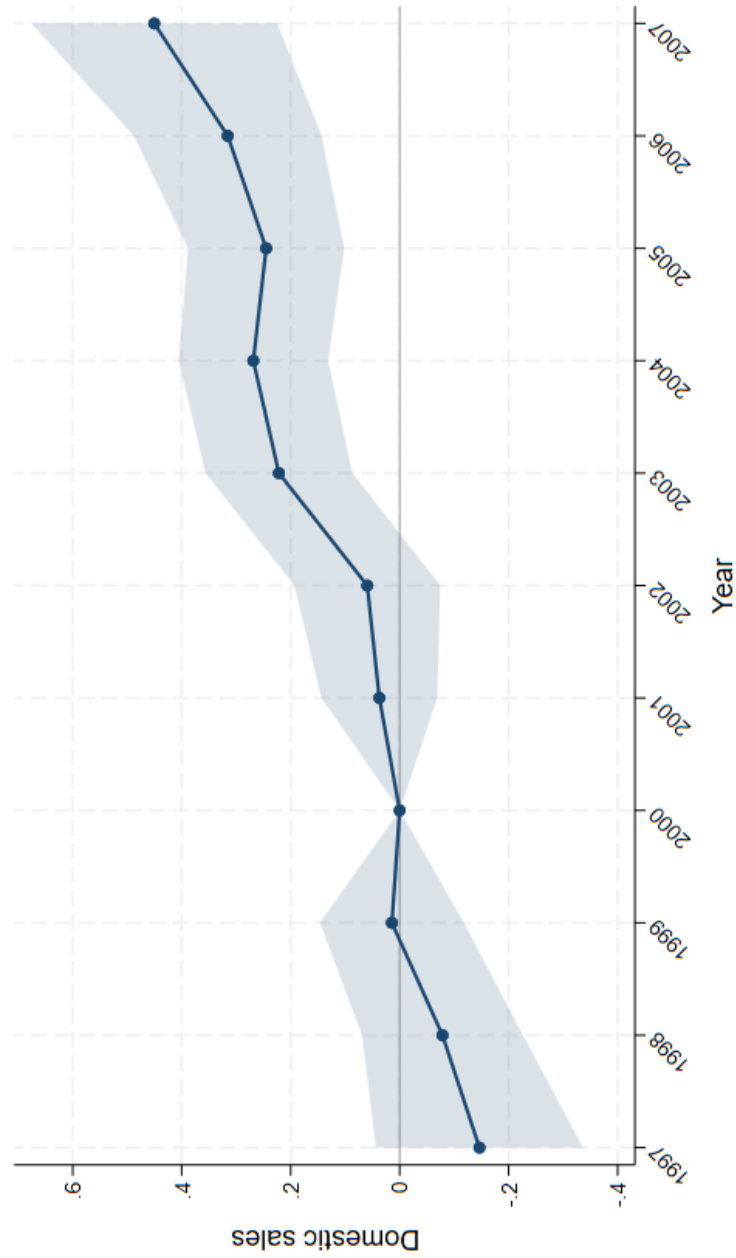
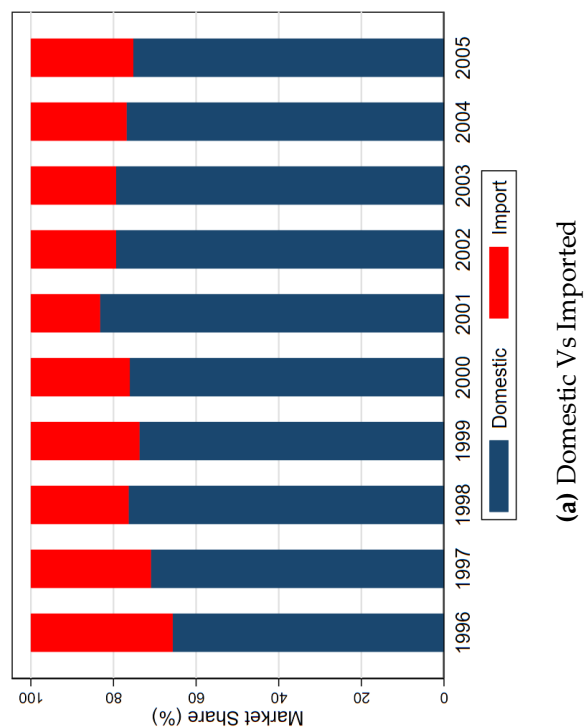
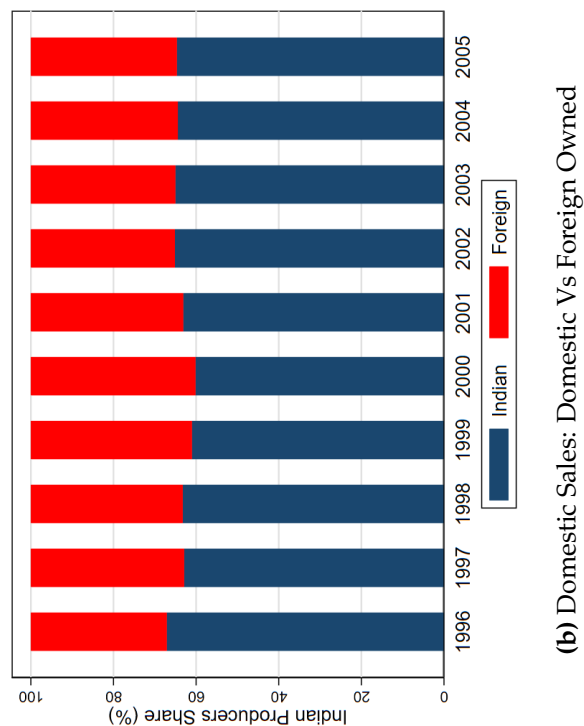


Figure D.7: TBTs and Its Impact on Domestic Sales (Non-exporters)

Notes: Figure plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in domestic sales between the firms which are affected by the automotive TBTs (regulations no. 84 and 104) and all other firms. All other firms include (i) firms which are treated with the other regulations, i.e., retail TBTs, and (ii) firms which are not affected by any TBT.



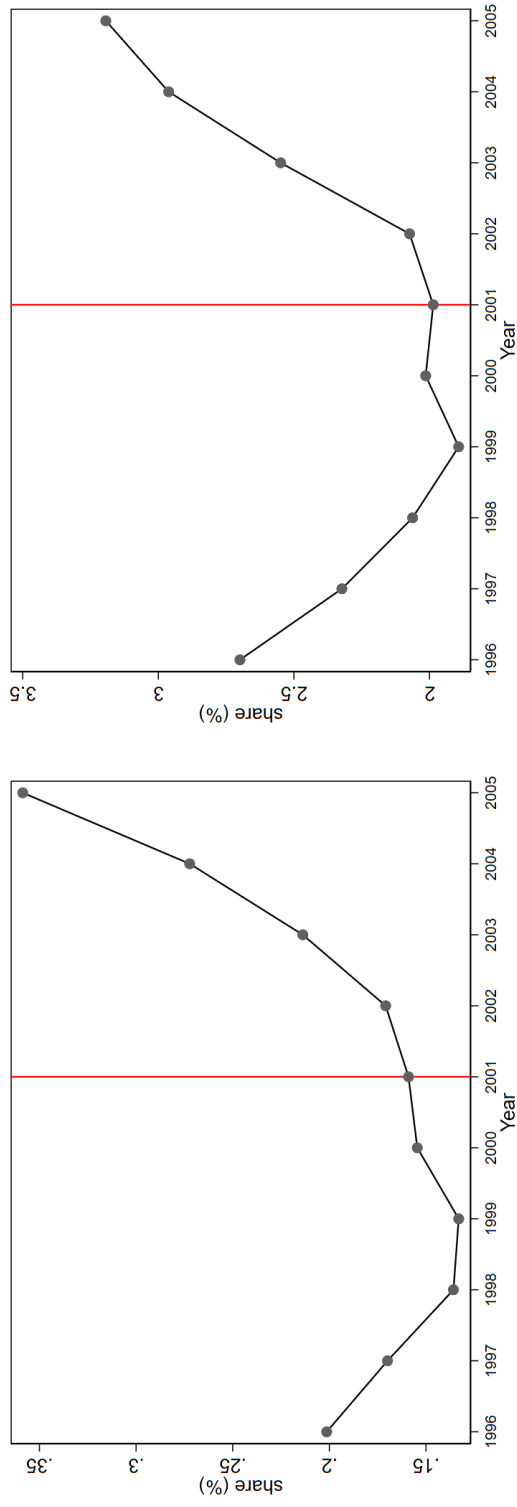
(a) Domestic Vs Imported



(b) Domestic Sales: Domestic Vs Foreign Owned

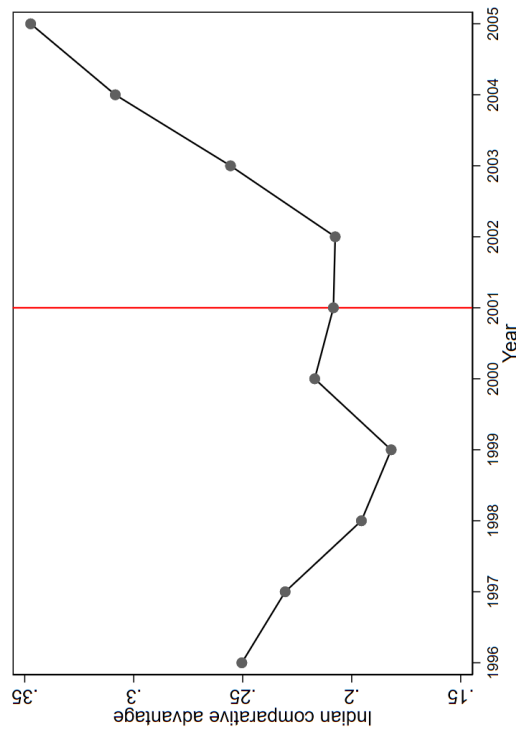
Figure D.8: Automotive Market Shares

Notes: Panel A of the Figure plots the yearly shares of domestic sales vs imports whereas Panel B plots the share of domestic sales between domestically-and foreign-owned firms.



(a) Share of Automotive Exports in Total Exports

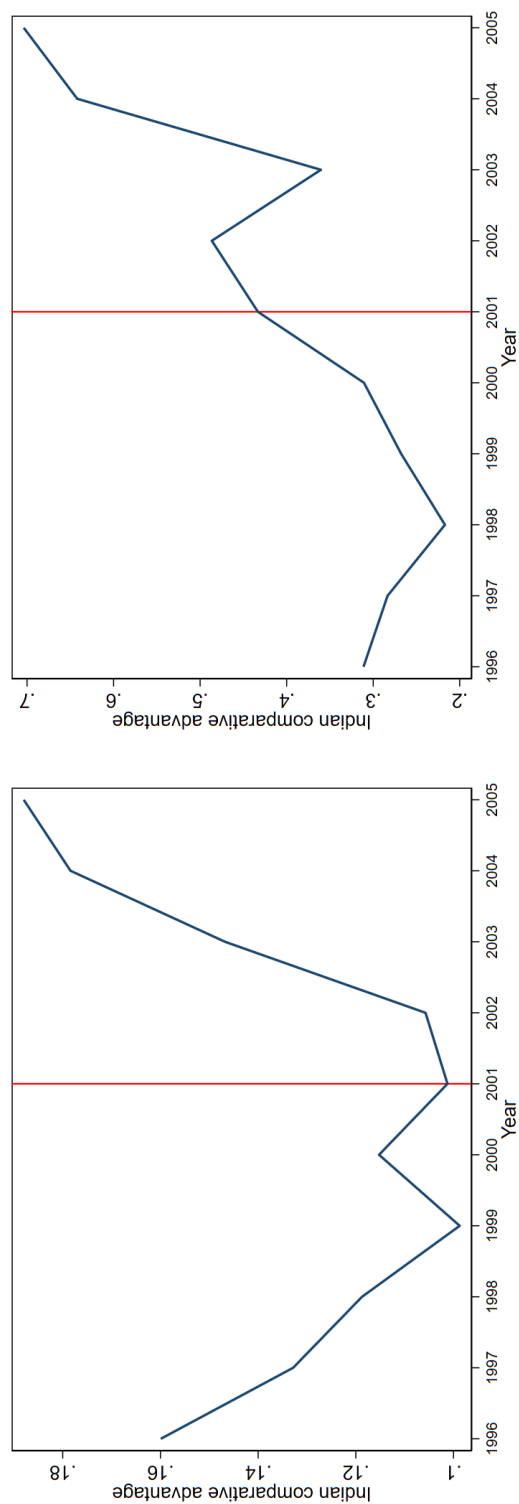
(b) Ratio of Automotive to All Other Industries' Exports



(c) Revealed Comparative Advantage of Automotive Exports

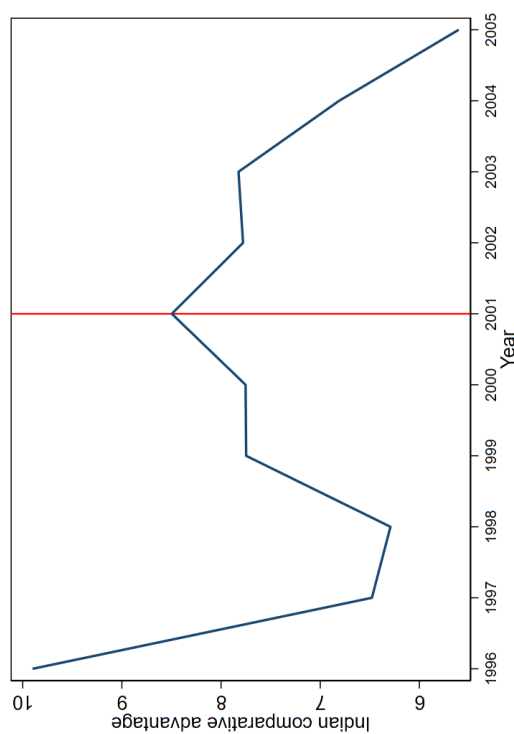
Figure D.9: Indian Automotive Exports

Notes: Panel A plots the Indian share of total world exports in the automotive sector, Panel B Figure plots the Indian automotive share of total Indian exports and Panel C Figure plots the Indian revealed comparative advantage (RCA) in automotive. The RCA index is computed as: $RCA = \frac{(X_{IND,i} / X_{IND,tot})}{(X_{WLD,i} / X_{WLD,tot})}$



(a) Advanced countries

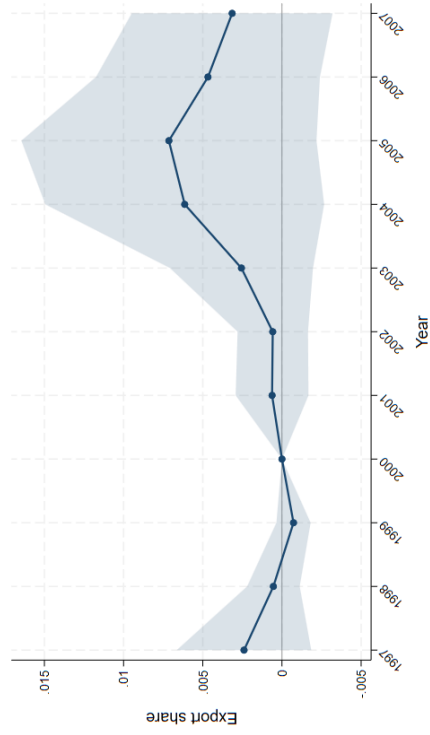
(b) Low-middle income countries



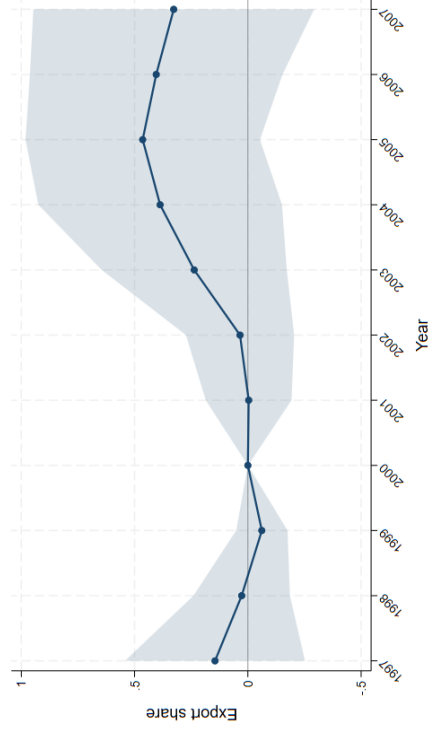
(c) Least developed countries

Figure D.10: Indian Comparative Advantage, By Destination Countries

Notes: RCA by country group is computed as $RCA = (X_{IND,i,g} / X_{IND,tot,g}) / (X_{WLD,i,g} / X_{WLD,tot,g})$, where g denotes the country group (advanced, middle-income, least-developed).



(a) Share of Indian Automotive exports – Using OLS



(b) Share of Indian Automotive exports – Using PPML

Figure D.11: Indian Automotive Exports

Notes: Panel A plots the yearly OLS coefficients (and their 95% confidence intervals) which are the differences in the shares of total Indian exports between the industries which are affected by the automotive TBTs (regulations no. 84 and 104) and all other industries. Panel B plots the yearly PPML coefficients. Year and industry fixed effects are accounted for and standard errors are clustered at industry (4 digits) level.